

Verifying Agent Repairs: Evidence Quality, Deployment Gains, and Cost

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Abstract

Self-improving agent loops change an agent’s harness (its prompts, tool descriptions and control code) and keep a change only when verification evidence says it helps. We measure what that evidence is worth by deploying every candidate change, accepted or rejected, on held-out tasks, and count its cost in agent episodes. In one round of attribution-driven repair of a DeepSeek-V4-Flash agent on τ^2 -bench, mostly of injected faults, no patch set chosen by up to ten verification policies beat applying the attributed patches unverified, in a test with little power. We then ran RRSI, a regularised self-improvement loop that edits harness code, for 20 rounds on three trajectories in each of two domains and on AppWorld, and scored seven kinds of evidence on the 183 candidates it measured, under registered analyses. Two findings are supported. Stopping each evaluation once RRSI’s rule is predicted to reject the candidate reproduced full re-evaluation’s decision in 93% of rounds at 57% of its episodes and was non-inferior to it at a margin of 0.25 points per round, which keeping the incumbent fails; it was also on AppWorld and on a trajectory with a stronger proposer. And evidence that never runs the changed agent on solved tasks accepted changes that crashed the harness or blocked legitimate actions. Which evidence predicts deployment gain remains open: full re-evaluation itself gained +0.04 points per round ($[-0.35, +0.43]$) on the registered trajectories, so fixed-budget samples and replay evidence, non-inferior to it at the registered margin of 0.75 points, are no more informative than keeping the incumbent, which passes that test too.

1 Introduction

A growing family of systems improves LLM agents from their own failures. The agent runs, an analyst model reads the failed trajectories, a proposer changes the agent’s *harness*, the prompts, tool descriptions and code around a frozen model, and

the loop keeps the change or discards it (Shinn et al., 2023; Zhao et al., 2024; Hu et al., 2025; Zhang et al., 2025a; Agrawal et al., 2026; Chen et al., 2026; Zhang et al., 2026b; Xia et al., 2026). The check that decides is where these systems differ most: some replay the failure with the change applied (Ma et al., 2025; Lin et al., 2026; Shah, 2026), some ask an LLM judge, some re-run tasks that used to pass (Chen et al., 2026), and RRSI re-evaluates every candidate twice on its whole training set (Xia et al., 2026). What every check is meant to estimate is a change’s *deployment gain*, how much it raises success on tasks the loop has not seen, and that gain can differ from what the loop observes on its own tasks: RRSI reports in-distribution gains that shrink out of distribution (Xia et al., 2026).

This paper asks two questions about verification evidence. *Which evidence predicts a change’s deployment gain?* And *is the evidence worth what it costs?* Answering the first needs ground truth that verification does not provide: deploying the changes, from the first step, on independent tasks. Answering the second needs an account of what verification costs and what the change it selects is worth over a deployment horizon, which we write as $\text{Net}(N)$ (§3). We price every piece of evidence in agent episodes and answer both questions in two studies.

Study 1 (§4, completed) is one round of attribution-driven repair of a DeepSeek-V4-Flash agent on the retail and airline domains of τ^2 -bench (Barres et al., 2025), with patches that rewrite policy sections, tool descriptions and the step limit. We record banks of targeted replays, null replays, judge checks and regression runs for 108 patches per domain, let eleven verification policies choose patch sets from them, and deploy the sets on held-out tasks. Applying the attributed patches without verification raised held-out success by 11.7 (retail) and 6.7 (airline) points, mostly on episodes with

injected faults, and no verified set did better in the primary comparison. Replay evidence ranked the deployed sets poorly, but the differences it predicted were below our deployment noise; judge checks cost 22–72% of a replay, and a fifth of unpatched replays succeed anyway.

Study 2 (§5, registered) tests evidence quality with the power and the modification space Study 1 lacked. We build on RRSI’s open-source loop (Xia et al., 2026), whose proposer edits harness code over many rounds of natural failures, port it to DeepSeek models and τ^2 -bench, make targeted replays work for code-level changes, and replace RRSI’s selection step with each kind of evidence at equal budget. We deploy every candidate and incumbent on held-out tasks, with endpoints, decision rule and analysis code fixed before any held-out episode was run. On a first trajectory per domain, evidence that does not run the changed agent accepted a change that crashed the harness. On two further registered trajectories, replays without regression runs accepted a change that blocked legitimate actions, and no rule was shown to beat keeping the incumbent. Full re-evaluation gained so little there that cheap evidence passing a non-inferiority test against it says little, because keeping the incumbent passes too. Sequential evaluation, which stops a candidate’s evaluation once RRSI’s rule is predicted to reject it, is the exception: it made full re-evaluation’s decision in 93% of rounds at 57% of its episodes and was non-inferior at a margin three times smaller, on AppWorld with a code agent and with a thinking proposer as well.

Our contributions are an evaluation protocol that scores verification decisions by deploying every candidate, accepted or rejected, with an explicit cost account; sequential verification, which keeps a loop’s decisions at about half the evaluation cost; and the measurements of both studies, released with the environments, banks and evidence. The loop, its regularised edits and its selection rule are RRSI’s (Xia et al., 2026); we change only the evidence its selection step uses. The multi-fidelity allocation method we started from, which assumed that judge checks are cheap, is reported as a design study in Appendix S.

2 Related Work

Self-improving agents and how they verify. Reflexion, Self-Refine and ExpeL turn feedback into verbal lessons (Shinn et al., 2023; Madaan

et al., 2023; Zhao et al., 2024); ProTeGi, OPRO, TextGrad and DSPy rewrite prompts (Pryzant et al., 2023; Yang et al., 2024; Yuksekogonul et al., 2025; Khattab et al., 2024); ADAS, the Darwin Gödel Machine and GEPA search over designs (Hu et al., 2025; Zhang et al., 2025a; Agrawal et al., 2026); ACE evolves a context playbook through incremental updates (Zhang et al., 2026b). Harness-Fix repairs harness code and promotes a bundle of repairs only if it solves more validation tasks while breaking at most two (Chen et al., 2026); Bayesian-Agent applies threshold rules to posteriors over skills (Wu et al., 2026). RRSI regularises harness self-improvement: it bounds and anneals the number of edits per candidate, screens benchmark-specific edits with a critic, prunes stale edits, and selects candidates by full re-evaluation against a calibrated noise floor (Xia et al., 2026). These systems report the success of the final agent; we evaluate the evidence behind each decision, and Study 2 uses RRSI’s code as its loop.

Failure attribution. Who&When and its extension show that locating the responsible agent and step is far from solved (Zhang et al., 2025b; Liu et al., 2026). AgenTracer trains an attribution model on counterfactual replays and injected faults (Zhang et al., 2026a); AgentDebug and ErrorProbe pair taxonomies or dependency tracing with corrective feedback (Zhu et al., 2025; Li et al., 2026); MAST and AdaMAST give attribution a vocabulary (Cemri et al., 2025, 2026).

Verification by intervention. DoVer validates hypotheses by editing a message or plan and re-running (Ma et al., 2025); REFLECT tests patches by targeted replay (Lin et al., 2026); CausalFlow and Causal Agent Replay score step interventions causally (Bonagiri et al., 2026; Shah, 2026); Agent-DebugX gates reruns by policy (Zhu et al., 2026). None of them measures whether its evidence predicts deployment or what the evidence costs. Program repair has long warned that a patch passing the failing test may not fix the fault (Smith et al., 2015).

3 Verification Evidence and Deployment Gain

Changes and their deployment gain. A loop holds an incumbent harness h and proposes a change c : a rewritten prompt section or tool description, a code edit, or a set of these. Its *de-*

ployment gain $\Delta p(c)$ is the difference in success rate between h with c and h alone, measured from the first step on tasks independent of everything the loop saw. A verification rule spends C_{verify} episodes on evidence, forms an estimate $\hat{\Delta}$ of the change’s effect and accepts the change if the estimate clears a threshold.

Evidence. We study the evidence that self-improvement loops use. *None* accepts the proposed change. An *LLM judge* reads the change and the failures it targets and predicts its effect, at a cost in tokens but no episodes; a *single-step check* regenerates the failing step with the change and asks a judge whether the error is gone. A *targeted replay* resumes a failed trajectory from the failing step with the change and succeeds with probability $b + (1 - b)q$, where q is the share of such failures the change repairs; a *null replay* does the same without the change and measures the *spurious recovery* rate b , without which replay success cannot be interpreted. A *regression run* is a previously solved task, from the first step, with the change; a *sample* or *full evaluation* runs candidate and incumbent on a random sample of, or all, training tasks. Replays and checks see only the failures a change targets; only regression runs and evaluations see harm to tasks that already succeed.

Is the evidence worth its cost? Compare a rule E with a reference rule, here applying changes without verification. Suppose E ’s choices change the deployed agent’s success by Δp and its running cost by Δc_{run} per episode, that E spends C_{verify} episodes, and that it changes the one-off cost of reviewing and maintaining the applied changes by ΔC_{change} . Over the next N deployed episodes E is worth

$$\text{Net}(N) = N(v\Delta p - \Delta c_{\text{run}}) - C_{\text{verify}} - \Delta C_{\text{change}}, \quad (1)$$

with every cost in episodes of the unmodified agent’s running cost and v the value of one success in the same unit. The one-off costs do not grow with N , so once Nv is large compared with C_{verify} the sign of $\text{Net}(N)$ is the sign of $v\Delta p - \Delta c_{\text{run}}$: evidence pays if the changes it selects are better, or cheaper to run. Its worth then rests on its *quality*, which we measure in two ways: how well its estimates correlate with, and are calibrated against, measured deployment gains, and the *decision value* of a rule that uses it, the mean deployment gain of the changes it accepts. Appendix A splits a patch’s

net effect into repair, spurious recovery and regressions.

4 Study 1: One Round of Attribution-Driven Repair

Agent and environments. The agent is DeepSeek-V4-Flash, with the provider’s default sampling, on the retail and airline domains of τ^2 -bench (Yao et al., 2024; Barres et al., 2025), which have an LLM user simulator and database-state rewards. Attribution, taxonomy induction (Wan et al., 2024), patch generation and judging use DeepSeek-V4-Pro. The patchable components are the sections of the domain policy, every tool description and the step limit: 24 components on retail and 21 on airline. Each training episode runs clean or with one injected fault written as a plausible authoring mistake (retail: a policy section allowing gift-card refunds, a cancellation rule without the reason check, a product-tool description that confuses item and product ids, a tight step limit; airline: a wrong bag allowance, a 7-day instead of 24-hour cancellation window, a missing basic-economy restriction, proactive compensation, a tight step limit). Clean episodes contribute natural failures, 22 of 137 on retail and 28 of 112 on airline. A third environment, ShopDesk, is described in Appendix M.

Patches and banks. The analyst groups failures into categories and blames components. For 36 cells (category, component) per domain, the blamed cells in order of blame plus random unblamed ones, it writes three patches each. For every patch we record 20 targeted replays from the attributed failing step, each paired with a single-step check; for every category, 20 null replays; and for every patch, 4 regression runs on solved training tasks, extended to 20 for the 121 patches that any policy accepted. An unpatched baseline of 10 runs on each of the 111 regression tasks gives task-matched estimates of harm. Policies run offline on these *intervention banks*, with every experiment priced in measured tokens relative to a full replay.

Policies and deployment. Eleven policies spend a budget of episodes from the banks and output a patch set (Appendix D). *Apply attributed* uses no evidence: for each category it applies the first patch of the most-blamed component and of any component with at least 40% of the blame. *Replay-each* accepts a hypothesis on two successes in three

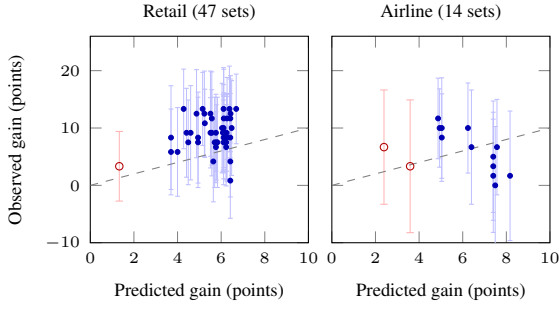


Figure 1: Each distinct deployed patch set’s gain over the unpatched agent as predicted from its bank replays (net of spurious recovery) against its observed held-out gain, with 90% intervals. Filled: sets containing the step-limit patch; open: sets without it. Dashed: $y = x$. Correlations are 0.23 (retail) and -0.49 (airline).

replays, as do-then-verify methods do (Ma et al., 2025); two HarnessFix-style gates require regression runs to break nothing, per patch or on the whole bundle (Chen et al., 2026); uncertainty sampling runs with and without cheap checks; four variants of CARVE, the multi-fidelity design we started from, and a net-effect allocation that charges regressions complete the list. We apply each policy’s budget-10 set, from five seeds and reduced to one patch per component, to unseen tasks: 40 retail tasks $\times$ 3 episodes (38 clean, 82 with an injected fault) and 20 airline tasks $\times$ 3 (25 clean), for five policies on airline. Retail was the development domain for every method.

4.1 Deployment gains

Table 1 reports absolute held-out success. Applying the attributed patches without verification raised success from 75.8% to 87.5% on retail (+11.7 points, 90% interval [4.5, 18.8]) and from 68.3% to 75.0% on airline (+6.7, [−3.3, 16.6]). No verified set did better: their differences from the unverified set range from -4.3 to $+0.3$ points on retail and from -4.0 to -0.3 on airline. This is not an equivalence result, and it concerns all held-out episodes; on clean episodes an exploratory comparison puts the bundle gate 4 points ahead, from seed sets identical to the unverified one (Appendix K). The intervals reach $+6.0$ points on retail and $+8.5$ on airline. Most of retail’s gain comes from fault-injected episodes: on its 38 clean episodes the unverified set gains $+5.3$ points [−5.1, 15.6], and on 158 clean episodes, 120 of them new, $+4$ [−3, 10] (Appendix K).

4.2 Does the evidence predict deployment?

Replays measure repair from the failing step, whereas deployed patches act from the first step and on every task. For every distinct deployed set we predict its gain over the unpatched agent from its patches’ replays as $\hat{F} \sum_i f_i [1 - \prod_k (1 - \hat{q}_k)]$, with $\hat{F}$ the held-out failure rate, f_i the category shares and $\hat{q}_k = (p_k - b_i)/(1 - b_i)$ the replay success p_k net of the category’s spurious recovery b_i . Figure 1 compares the predictions with the observed gains. They correlate weakly, at $r = 0.23$ over 47 distinct retail sets and -0.49 over 14 airline sets; predictions from the judge checks, calibrated to replay success, do no better ($r = 0.14$ and -0.05). In level, replays predict airline’s gains (6.0 points on average against 6.1 observed) but not retail’s (5.6 against 9.1).

The ranking test has little power, for a reason that matters beyond this study. Replay predictions differ between sets by a standard deviation of 1.0 point on retail and 1.7 on airline, whereas a set’s held-out success measured on 120 (60) episodes has a run-to-run standard error of 2.6 (3.8) points; the observed gains spread by 2.8 (3.7), no more than that noise. A perfect predictor of differences this small would be indistinguishable from a useless one. Testing whether evidence predicts deployment needs candidates that differ by more than the deployment noise and more precise deployment measurements; Study 2 is designed for both.

4.3 What the evidence costs

Table 4 (Appendix B) collects the measurements. *Judge checks are not cheap.* A single-step check costs 22–72% of a full replay in tokens, and its information per token is at best similar to a replay’s: against a patch’s replay success rate its efficiency ratio is 0.88 [0.46, 1.53] on retail and 1.25 [0.19, 3.95] on airline, and against single replays it is below 1 for every judge (Appendix I). *Replays need controls:* without any patch, 20–23% of replays from the failing step succeed anyway, and the rate ranges from 0 to 0.5 across categories. *Harm is the least measurable quantity:* the mean effect of a patch on previously solved tasks is -1.0 points on retail and -0.6 on airline, with 90% intervals of about ± 2.5 , and patches differ by a spread of 2–3 points; detecting a 3-point regression against a 15% base rate with 90% power takes about 1,200 runs per patch (Proposition 2).

Patch set (budget 10)	Retail (40 tasks $\times$ 3)					Airline (20 tasks $\times$ 3)				
	All	Clean	−Apply [90%]	#p	C	All	Clean	−Apply [90%]	#p	C
Unpatched	75.8	81.6		0.0		68.3	76.0		0.0	
Apply attributed	87.5	86.8	−	7.0	0	75.0	76.0	−	5.0	0
Replay-each	83.5	86.8	−4.0 [−9.8, 1.8]	6.0	10					
HarnessFix-style, per patch	83.2	90.0	−4.3 [−12.8, 4.2]	0.8	9	71.0	76.8	−4.0 [−11.9, 3.9]	1.0	8
HarnessFix-style, bundle	84.8	86.8	−2.7 [−5.0, −0.4]	5.6	6	71.7	76.8	−3.3 [−9.3, 2.6]	3.0	6
Uncertainty sampling	85.8	86.3	−1.7 [−6.9, 3.5]	5.8	10					
Uncertainty + checks	84.7	82.1	−2.8 [−8.4, 2.7]	5.8	10					
CARVE, replay only	87.8	86.8	0.3 [−5.3, 6.0]	6.0	10					
CARVE	83.8	80.5	−3.7 [−9.9, 2.6]	5.2	10	74.3	72.0	−0.7 [−8.8, 7.5]	4.2	10
CARVE, measured check	84.8	86.3	−2.7 [−7.6, 2.3]	6.0	10					
CARVE, robust check	86.3	83.7	−1.2 [−6.3, 4.0]	6.0	9					
Net-effect	83.2	83.2	−4.3 [−9.9, 1.3]	3.4	10	74.7	76.0	−0.3 [−9.1, 8.5]	3.6	10

Table 1: Held-out success (%) of the patch set each policy selects with a budget of 10 episodes, mean over five policy seeds. *All*: every held-out episode (retail: 82 of 120 with an injected fault; airline: 35 of 60). *Clean*: episodes without an injected fault (38 and 25). *−Apply*: paired difference from Apply attributed over all episodes, in points, with a task-clustered 90% interval. *#p*: patches deployed; *C*: verification episodes spent.

4.4 Is verification worth its cost?

Table 5 (Appendix B) evaluates Eq. 1 for every verified set against applying the attributed patches. The verification bill is at most 10 episodes, about two cents of τ^2 compute at our measured cost, while at $N = 1000$ the 90% intervals of $\text{Net}(N)$ span 370–520 episodes at $v = 1$, mostly from the uncertain running cost, and 630–1,830 at $v = 10$. The sign of $\text{Net}(N)$ is thus decided by Δp and the running cost, not by the bill, and every interval includes zero. In this regime the question “is verification worth its cost” reduces to “does it select better changes”, which Study 1 could not resolve; scored on the banks, the verifying policies were not reliably better either (Appendix D).

5 Study 2: Verification Evidence Inside RRSI

Study 1 patches only text, runs one round, draws most failures from injected faults, and compares sets whose predicted differences are below its deployment noise. Study 2 builds on RRSI (Xia et al., 2026), whose proposer edits harness code over many rounds of natural failures. We run RRSI’s loop, collect every kind of evidence for every candidate it measures, deploy every candidate, accepted or rejected, and ask what each kind of evidence would have decided. Appendix N describes the port, the registrations and every change made after them.

Loop. We vendor RRSI’s code (Apache 2.0) and change only its model backend, its cost accounting and one rule for empty proposals. Analyst, pro-

poser, critic and judge are DeepSeek-V4-Pro with thinking disabled; the agent and user simulator are DeepSeek-V4-Flash at temperature 0, at which our episode loop matches τ^2 ’s own orchestrator; the starting harness is τ^2 ’s reference agent. Each round the proposer drafts $m = 2$ candidates under an edit budget that anneals from 4 to 1, a critic rejects edits that hard-code task data or touch the grader, and RRSI evaluates the candidates that pass a smoke test twice on every evolve task. It accepts a candidate that beats the incumbent by more than a noise band δ , calibrated from three evaluations of the base harness, or that is within the band and cheaper to run or of an untried kind, and it rejects a harness that raises errors in more than 2% of trials (Appendix N). Evolve tasks are τ^2 ’s training split (74 retail, 30 airline) and held-out tasks its test split (40 and 20). The loop ran $T = 20$ rounds per domain on three independent trajectories, r1–r3, which differ only in the samples of the proposer, analyst and critic, with r2 and r3 started after r1’s results were known (below).

Evidence and decisions. For each measured candidate we collect the evidence of §3 as estimates of its change ΔS in evolve success (Table 19, Appendix N): an LLM judge’s prediction from the diff and the failures it targets (no episodes); *sample@b*, the paired difference on b evolve episodes; *replay@b*, the failure share times the success rate of b targeted replays of failed evolve trials; *replay−null@b*, which subtracts the success rate of null replays with the incumbent; and *net@b*, which splits b between both kinds of replay

and regression runs on solved tasks, with or without a Beta prior on the repair rate. Budgets of 10, 40 and 80 episodes per candidate are subsamples of one collection, four replays with each candidate and four null replays of each failed evolve task (at most 20 per round), plus RRSI’s own full evaluation. A replay re-executes the recorded prefix through the new harness and goes live where it would first act differently. Every rule decides with RRSI’s rule, with ΔS replaced by its estimate and δ by twice the estimate’s standard error; full evaluation keeps RRSI’s calibrated δ . The rules thus differ in their evidence and in the band it implies, which decides when RRSI’s within-band branch applies; Appendix Q decides every rule with one criterion.

Ground truth and estimand. Every incumbent and measured candidate is deployed from the first step on the held-out tasks, 240 episodes each (retail 40 tasks $\times$ 6, airline 20 $\times$ 12), and the base and final harnesses on 400 and 240. A candidate’s deployment gain $\Delta p(c \mid H_t)$ is its held-out success minus that of the incumbent H_t it was proposed against, paired by task. A rule E ’s *decision value* on a set $\mathcal{T}$ of rounds of both domains is $D(E) = \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} \sum_c P_E(c \mid t) \Delta p(c \mid H_t)$, the mean expected deployment gain of its choice per round (zero when it keeps H_t), with P_E taken over 400 subsamples of the evidence. D is conditional on RRSI’s trajectories: it scores each rule’s decisions in the states RRSI visited, not a loop driven by that rule, whose later incumbents and candidates would differ. For r1 the registered endpoints are (P1) *transfer*, the held-out gain of the final harness over the base, and (P2) D at $b = 40$ with three Holm-adjusted contrasts (Appendix O).

Further trajectories. Because r1’s values rested on one trajectory per domain and its registered analysis gave RRSI’s error check to full evaluation only (§5.1), we registered r2 and r3 before any of their episodes ran, with the primary analysis of this study: D over their 80 rounds with the error check applied to every rule’s own episodes (the *common error check*), and one-sided non-inferiority tests of sample@40, net@40 and replay–null@40 against full evaluation, margin 0.75 points per round, Holm-adjusted at $\alpha = 0.05$. A bootstrap resamples, within domain and trajectory, blocks of rounds that share an incumbent, and held-out tasks ($B = 2000$). Two sequential rules (Appendix N) form a second family. r2 and r3 use corrected com-

ponent labels and are otherwise as r1. One more registered trajectory runs the same loop and analysis on AppWorld (Trivedi et al., 2024), whose agent writes Python against simulated apps (E3, Appendix P).

5.1 Results

The loop. Over r1–r3 RRSI drafted 240 candidates and measured 154 (the critic rejected 47, 31 drafts made no change, 8 failed the smoke test); 106 of the 154 changed harness code. It accepted 5 candidates on r1, all in the first five rounds, and 21 on r2 and r3, as late as round 19; 19 of the 26 were within the noise band. Candidates’ mean deployment gain was -1.5 , -1.25 and $+0.14$ points on r1–r3, and 63 of 154 were positive.

Transfer (P1). Airline’s evolve gains transferred on r1 and r2, by $+8.75$ [$+2.5, +15.8$] and $+7.9$ [$+2.5, +14.2$] points against $+13.3$ on evolve tasks. On r3 its six acceptances, all within the band, left evolve success unchanged ($+1.7$ [$-2.5, +5.8$] held out). Retail’s evolve gains of 4.1, 4.7 and 7.4 points never transferred: -0.25 [$-3.0, +2.75$], 0.0 [$-2.0, +1.75$] and -4.25 [$-7.5, -1.0$].

First trajectory. On r1, most of the losses of no verification, the judge and replay without null replays come from one candidate (retail, round 6), a check that reads an attribute the harness never sets: the harness raised errors in half of its evolve episodes and held-out success fell by 51.7 points, yet the critic passed it and the judge predicted a gain. In the registered analysis, which gave RRSI’s error check to full evaluation only, full evaluation led with $+0.22$ points per round [$-0.06, +0.62$], and net@40 was neither shown non-inferior nor inferior to it ($p = 0.058$, Holm 0.17; Appendix O). With the common error check, accepting whatever passes the check still loses 0.58 points per round.

Primary analysis (r2 and r3). On their 80 rounds, full evaluation’s decision value was $+0.04$ points per round [$-0.35, +0.43$] at 141 episodes (Table 2). Sample@40 and net@40, at 54 and 49 episodes, were non-inferior to it: differences -0.10 [$-0.57, +0.35$] and -0.14 [$-0.58, +0.27$] (one-sided $p = 0.018$ and 0.015 , Holm 0.045 each). Replay–null@40 was not (-0.43 [$-0.94, +0.05$], $p = 0.14$); its own value, -0.39 [$-0.79, -0.03$], lies below zero, 40% of it from one candidate (r3 retail, round 11). That candidate held every write whose arguments it could not trace to earlier tool

Evidence	b	Decision value D [90%], common error check			r2+r3			
		r1 (40 rounds)	r2+r3 (80, primary)	r1-r3 (120)	Accepts	Agree	Ep.	Net(10^4)
Keep incumbent	0	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	0.00	0.74	0	0
None	0	-1.87 [-3.52, +1.53]	-0.47 [-1.52, +0.62]	-0.94 [-1.94, +0.22]	0.91	0.25	0	-818
LLM judge	0	-1.72 [-3.30, +1.37]	-0.40 [-1.41, +0.67]	-0.84 [-1.75, +0.27]	0.81	0.30	0	-731
None + check	10	-0.58 [-1.88, +1.56]	-0.47 [-1.52, +0.62]	-0.51 [-1.39, +0.46]	0.91	0.25	13	-831
Judge + check	10	-0.43 [-1.53, +1.41]	-0.40 [-1.41, +0.67]	-0.41 [-1.20, +0.47]	0.81	0.30	13	-744
Full (RRSI)	all	+0.22 [-0.04, +1.11]	+0.04 [-0.35, +0.43]	+0.10 [-0.18, +0.41]	0.26	1.00	141	-108
Sequential full	$\leq$ all	+0.15 [-0.03, +0.79]	+0.02 [-0.34, +0.35]	+0.06 [-0.18, +0.32]	0.20	0.93	81	-71
Sequential sample	$\leq$ 80	+0.12 [-0.15, +0.98]	+0.07 [-0.26, +0.41]	+0.09 [-0.15, +0.38]	0.23	0.78	53	10
Sample	10	-0.04 [-0.42, +0.77]	-0.08 [-0.54, +0.37]	-0.07 [-0.39, +0.30]	0.39	0.60	13	-107
Replay	10	-0.37 [-1.26, +0.88]	-0.67 [-1.56, +0.17]	-0.57 [-1.20, +0.13]	0.63	0.44	13	-1,000
Replay-null	10	-0.05 [-0.25, +0.16]	-0.36 [-0.74, -0.01]	-0.25 [-0.53, -0.00]	0.18	0.69	11	-594
Net	10	-0.07 [-0.54, +0.93]	-0.17 [-0.73, +0.35]	-0.13 [-0.52, +0.30]	0.48	0.56	12	-217
Sample	40	+0.04 [-0.35, +0.93]	-0.06 [-0.58, +0.45]	-0.02 [-0.37, +0.38]	0.40	0.67	54	-119
Replay	40	-0.40 [-1.56, +1.34]	-0.74 [-1.83, +0.36]	-0.63 [-1.46, +0.30]	0.86	0.33	50	-1,126
Replay-null	40	-0.07 [-0.34, +0.21]	-0.39 [-0.79, -0.03]	-0.29 [-0.58, -0.01]	0.20	0.68	45	-674
Net	40	-0.04 [-0.53, +0.76]	-0.10 [-0.62, +0.39]	-0.08 [-0.45, +0.31]	0.44	0.63	49	-152
Sample	80	+0.08 [-0.30, +1.06]	+0.04 [-0.46, +0.57]	+0.06 [-0.30, +0.47]	0.38	0.78	94	-57
Replay	80	-0.40 [-1.55, +1.34]	-0.71 [-1.80, +0.42]	-0.60 [-1.45, +0.33]	0.88	0.31	56	-1,101
Replay-null	80	-0.12 [-0.60, +0.45]	-0.51 [-1.03, -0.05]	-0.38 [-0.74, -0.02]	0.25	0.65	84	-848
Net	80	-0.02 [-0.53, +0.81]	-0.04 [-0.55, +0.45]	-0.03 [-0.42, +0.36]	0.41	0.69	98	-123

Table 2: Study 2: decision value D of each rule with the error check applied to every rule’s own episodes, the held-out gain of its choice over the incumbent in points per round (keeping the incumbent scores 0), with 90% intervals from a bootstrap of blocks of rounds that share an incumbent and of held-out tasks, within domain and trajectory. r1: the first trajectory per domain (its registered analysis is in Table 20); r2+r3: the registered primary analysis; r1-r3: pooled. + *check* rules run 10 episodes of each candidate only for the check; sequential rules stop a candidate’s episodes early. Accepts (mean probability of accepting a candidate in a round), Agree (share of rounds in which the rule makes full evaluation’s choice), evidence episodes per round, and Net(10^4) (Eq. 1 against keeping the incumbent, $N = 10^4$ deployed episodes, a success worth $v = 10$ episodes, in episodes) are for r2+r3. Full: RRSI’s own decision. Best value per column in bold.

results; it raised no harness errors but blocked legitimate writes, lowering held-out success by 12.5 points. Full evaluation, sample and net, which run solved tasks, rejected it; replay-null, which replays only failures, and the judge accepted it. Full evaluation’s advantage over accepting every candidate, +0.51 [-0.51, +1.48], was not resolved: no rule was shown to beat keeping the incumbent. Because full evaluation gained so little, keeping the incumbent is itself non-inferior to it at the registered margin ($p = 0.004$), more clearly than sample@40 and net@40; at a margin of 0.25 points neither they nor keeping the incumbent are ($p \geq 0.17$; post hoc, Table 22). Their non-inferiority therefore shows that they lose little against a rule that gained little, not that they preserve a useful improvement process.

Sequential evaluation. Sequential full evaluation was non-inferior to full evaluation, -0.02 [-0.12, +0.07] at 81 episodes per round (57%), as was sequential sampling capped at 80 episodes,

+0.03 [-0.26, +0.35] at 53 (both $p < 0.001$). Unlike the fixed-budget rules, sequential full evaluation stays non-inferior at a margin of 0.25 points ($p < 0.001$) and makes full evaluation’s choice in 93% of rounds, against 74% for keeping the incumbent; sequential sampling makes it in 78% and is non-inferior at 0.5 points ($p = 0.002$). On r1, where they were designed, sequential full evaluation gained +0.15 at 65 episodes per round against +0.22 at 123 (agreement 96%).

All three trajectories. Pooled over 120 rounds, full evaluation gained +0.10 [-0.18, +0.41], with the same non-inferiority results (Holm $p \leq 0.018$; replay-null@40 $p = 0.058$). Full evaluation’s value was +0.22, +0.16 and -0.07 on r1-r3, and accepting every candidate’s -1.87, -1.08 and +0.14: trajectories differed more than rules.

Evidence or decision rule? Deciding full evaluation with twice its standard error, like every other rule, changes its value on r2 and r3 from

+0.04 to +0.03 and leaves the order of the evidence unchanged. Without the within-band branch every episode-based rule accepts in at most 3% of rounds and gains 0.00 to +0.10 points, except replay-null@40, which accepts in 7% and stays below full evaluation ($-0.21 [-0.47, -0.01]$; Table 23). The value and risk of RRSI’s decisions lie mostly in its within-band acceptances, 19 of 26.

A stronger proposer (r4). With thinking enabled, one airline trajectory of ten rounds measured six candidates, all helpful (descriptive, Appendix N). Full evaluation gained +0.92 points per round, so keeping the incumbent is no longer non-inferior to it ($p = 0.56$ at the registered margin), while sequential full evaluation made its choice in 95% of rounds and was non-inferior at 0.25 points ($p = 0.023$).

A code agent (E3). On AppWorld the 29 measured candidates barely changed held-out success (mean -0.2 points), and the final harness’s evolve gain of 4.2 points did not transfer ($0.0 [-2.1, +2.1]$). Full evaluation’s value was +0.03 points per round $[-0.30, +0.71]$ at 174 episodes (Table 21). The @40 rules came within 0.11 points of it but missed non-inferiority after Holm adjustment (one-sided $p = 0.018-0.025$, Holm 0.053); sequential full evaluation was non-inferior at 62% of its episodes ($p < 0.001$, also at a margin of 0.1 points) and made full evaluation’s choice in 98% of rounds.

Resolution. Few candidates’ gains exceeded the single-candidate detectable effect of 5.3–5.9 points (4–11%); differences between full evaluation and the @40 rules are detected with 80% power from 0.64–0.75 points per round, and correlations of the evidence with deployment gain were at most 0.33 (Appendix O).

Cost. Full evaluation cost 123 (r1) and 141 (r2, r3) episodes per round and the @40 rules about a third of that; weighting tokens by list prices changes no ranking. On r2 and r3, $\text{Net}(10^4)$ at $v = 10$ against keeping the incumbent was -108 episodes for full evaluation and positive only for sequential sampling, $+10 [-315, +350]$, which shows no economic advantage. Candidate evaluation was 24–54% of the loop’s own dollar cost on τ^2 , the proposal side’s model calls most of the rest, so sequential full evaluation would have cut the whole loop’s cost by 19% (7–32% per domain and

trajectory; 17% on AppWorld; Table 25). r1–r3 and E3 used about 83,000 episodes and \$168.

6 Discussion

What the evidence shows. Our results identify failure modes of verification and a cheaper way to reach the same decisions, not a ranking of evidence by predictive quality. Most changes did not help held out, so verification earned its place by rejecting harmful ones, and only evidence that runs the changed agent on solved tasks rejected both kinds we saw.

Resolution limited these studies more than price.

A round of full evaluation costs about 25 cents of τ^2 compute. What limited both studies was the resolution of deployment measurements: comparing evidence needs per-round differences of about 0.7 points or candidate gains above 3–5 points, and few changes reached that. Non-inferiority at a margin of 0.75 points says that cheap evidence loses little against a rule that itself gained little; keeping the incumbent passes the same test, so a comparison of verification rules needs a reference that gains, a margin below what doing nothing loses, or agreement with the reference’s decisions, the measure on which sequential evaluation stands apart.

Implications for self-improvement loops.

Loops that report only the final agent’s held-out success (Chen et al., 2026; Zhang et al., 2026b; Xia et al., 2026) measure all their decisions at once; on retail and AppWorld, evolve gains of 4–7 points never transferred. Recording the held-out gain of accepted and rejected changes shows which evidence earns its place. Replays need null replays and regression runs. Stopping an evaluation once rejection is predicted with high probability cut its episodes by 40–50% on both benchmarks and the loop’s whole cost by about a fifth; accepted candidates still complete their evaluation, so the traces the next round analyses are unchanged.

7 Conclusion

Inside RRSI’s loop most changes did not raise held-out success, and evidence that never runs the changed agent on solved tasks accepted changes that crashed the harness or blocked legitimate actions. Sequential evaluation reproduced full re-evaluation’s decisions in 93–98% of rounds at about half its episodes. Which evidence predicts deployment gain remains open.

Limitations

Both studies use one model family, with DeepSeek-V4-Flash as agent and DeepSeek-V4-Pro as analyst, proposer and judge, and two customer-service domains of τ^2 -bench; Study 1 adds ShopDesk and Study 2 one AppWorld trajectory. Study 1’s patches rewrite only policy sections, tool descriptions and a step limit; it runs a single round, mostly on injected faults (natural failures are 16–25% of training failures), and deploys patch sets only at budget 10, on 120 (retail) and 60 (airline) held-out episodes, where most of retail’s gain comes from fault-injected episodes. Its bank analyses rest on a harm model fitted to at most 20 regression runs per patch, and its edge priors use an analyst accuracy of 0.7 unless stated. Study 1’s agent used the provider’s default sampling and Study 2’s uses temperature 0. Study 2’s proposer is likely weaker than the model RRSI was developed with; a fourth trajectory with a thinking proposer was cut short because its drafts took over an hour and mostly ended without a proposal; on airline all six of its measured candidates helped, which suggests that a stronger proposer leaves less for verification to decide, but rests on one trajectory (Appendix N). Its held-out sets are τ^2 ’s test splits of 40 and 20 tasks, on which most candidates’ gains were not resolved. Its decision values are conditional on RRSI’s trajectories: they score each rule’s decisions in the states RRSI visited, not loops driven by those rules, which we did not run. A loop driven by sequential evaluation follows RRSI’s path until its first different decision, which the agreement rates make rare, except that RRSI shows the proposer the measured gains of rejected candidates, which would be early-stopped estimates; an immediately neutral acceptance may also change later exploration, which one-step decision values do not capture. $\text{Net}(N)$ sets the change cost to zero; charging each accepted change a fixed review cost favours rules that accept less and leaves the order of the sequential rules and full evaluation unchanged (Appendix Q). Three trajectories per domain show that between-trajectory variation is as large as the differences between rules, and the pooled analysis treats trajectories as fixed. The non-inferiority margin of 0.75 points per round was set before any result, but it is large relative to the decision values we observed; the smaller margins, the agreement rates and the decision-rule ablation are post hoc. On r1, the registered analysis gave RRSI’s error check to full evaluation only,

so the comparison with a common check is post hoc there; its held-out count was raised after six rounds, when some held-out episodes had run and a few results had been seen (Appendix N), and the registered count gives the same conclusions. While r2 and r3 ran, a few of their held-out success rates appeared in progress logs before analysis; the analysis code had been fixed and tested on r1 only. The sequential rules were designed on r1’s rounds and confirmed on r2 and r3. r1’s component labels did not recognise our harness’s files, which may have changed what its proposer was told to explore; r2 and r3 use corrected labels, so the trajectories differ in that respect as well as in sampling. Token prices for the models we used change; we report costs in episodes, and weighting tokens by list prices changes no ranking.

Ethics Statement

Self-improving agents change their own behaviour without a person reviewing each change. In Study 2 most proposed changes did not raise held-out success; one that crashed the harness in half of its episodes, and another that blocked legitimate writes, passed both an LLM critic and an LLM judge. That Study 1 detected no gain from verification does not show that verification is unnecessary; our measurements cannot exclude small harms per patch and should not be read as licence to skip review. On ShopDesk a single patch silently broke a quarter of tasks (Appendix M). Our environments use synthetic customer data only.

References

- Lakshya A. Agrawal, Shangyin Tan, Dilara Soyly, Noah Ziemis, Rishi Khare, Krista Opsahl-Ong, Arnab Singhvi, Herumb Shandilya, Michael J. Ryan, Meng Jiang, Christopher Potts, Koushik Sen, Alexandros G. Dimakis, Ion Stoica, Dan Klein, Matei Zaharia, and Omar Khattab. 2026. GEPA: Reflective prompt evolution can outperform reinforcement learning. In *International Conference on Learning Representations*.
- Victor Barres, Honghua Dong, Soham Ray, Xujie Si, and Karthik Narasimhan. 2025. τ^2 -bench: Evaluating conversational agents in a dual-control environment. *arXiv preprint arXiv:2506.07982*.
- Akash Bonagiri, Devang Borkar, Gerard Janno Anderias, Setareh Rafatirad, and Houman Homayoun. 2026. CausalFlow: Causal attribution and counterfactual repair for LLM agent failures. *arXiv preprint arXiv:2605.25338*.

768	Mert Cemri, Andrei Cojocaru, Melissa Pan, Shu Liu,	Aman Madaan, Niket Tandon, Prakhar Gupta, Skyler	822
769	Shubham Agarwal, Alexander Krentsel, Jay Tang,	Hallinan, Luyu Gao, Sarah Wiegrefe, Uri Alon,	823
770	Kannan Ramchandran, Joseph E. Gonzalez, Matei	Nouha Dziri, Shrimai Prabhumoye, Yiming Yang,	824
771	Zaharia, Alexandros G. Dimakis, and Ion Stoica.	Shashank Gupta, Bodhisattwa Prasad Majumder,	825
772	2026. Fantastic adaptive taxonomies and how to	Katherine Hermann, Sean Welleck, Amir Yazdan-	826
773	use them. <i>arXiv preprint arXiv:2607.16387</i> .	bakhsh, and Peter Clark. 2023. Self-refine: Iterative	827
		refinement with self-feedback. In <i>Advances in Neu-</i>	828
774	Mert Cemri, Melissa Z. Pan, Shuyi Yang, Lakshya A.	ral Information Processing Systems.	829
775	Agrawal, Bhavya Chopra, Rishabh Tiwari, Kurt		
776	Keutzer, Aditya Parameswaran, Dan Klein, Kannan	Reid Pryzant, Dan Iter, Jerry Li, Yin Tat Lee, Chen-	830
777	Ramchandran, Matei Zaharia, Joseph E. Gonzalez,	guang Zhu, and Michael Zeng. 2023. Automatic	831
778	and Ion Stoica. 2025. Why do multi-agent LLM	prompt optimization with “gradient descent” and	832
779	systems fail? <i>arXiv preprint arXiv:2503.13657</i> .	beam search. In <i>Proceedings of the 2023 Confer-</i>	833
		<i>ence on Empirical Methods in Natural Language</i>	834
780	Kathryn Chaloner and Isabella Verdinelli. 1995.	<i>Processing</i> .	835
781	Bayesian experimental design: A review. <i>Statisti-</i>		
782	<i>cal Science</i> , 10(3):273–304.	Jaineet Shah. 2026. Causal agent replay: Counterfactual	836
		attribution for LLM-agent failures. <i>arXiv preprint</i>	837
783	Mengzhuo Chen, Junjie Wang, Zhe Liu, Yawen Wang,	<i>arXiv:2606.08275</i> .	838
784	Haiming Zheng, and Qing Wang. 2026. From		
785	failed trajectories to reliable LLM agents: Diag-	Noah Shinn, Federico Cassano, Ashwin Gopinath,	839
786	nosng and repairing harness flaws. <i>arXiv preprint</i>	Karthik Narasimhan, and Shunyu Yao. 2023. Re-	840
787	<i>arXiv:2606.06324</i> .	flexion: Language agents with verbal reinforcement	841
		learning. In <i>Advances in Neural Information Pro-</i>	842
788	Peter I. Frazier, Warren B. Powell, and Savas Dayanik.	<i>cessing Systems</i> .	843
789	2008. A knowledge-gradient policy for sequential		
790	information collection. <i>SIAM Journal on Control</i>	Edward K. Smith, Earl T. Barr, Claire Le Goues, and	844
791	<i>and Optimization</i> , 47(5):2410–2439.	Yuriy Brun. 2015. Is the cure worse than the disease?	845
		overfitting in automated program repair. In <i>Proceed-</i>	846
792	Shengran Hu, Cong Lu, and Jeff Clune. 2025. Auto-	<i>ings of the 10th Joint Meeting on Foundations of</i>	847
793	mated design of agentic systems. In <i>International</i>	<i>Software Engineering</i> .	848
794	<i>Conference on Learning Representations</i> .		
795	Omar Khattab, Arnav Singhvi, Paridhi Maheshwari,	Harsh Trivedi, Tushar Khot, Mareike Hartmann, Ruskin	849
796	Zhiyuan Zhang, Keshav Santhanam, Sri Vard-	Manku, Vinty Dong, Edward Li, Shashank Gupta,	850
797	hamanan, Saiful Haq, Ashutosh Sharma, Thomas T.	Ashish Sabharwal, and Niranjan Balasubramanian.	851
798	Joshi, Hanna Moazam, Heather Miller, Matei Zaharia,	2024. AppWorld: A controllable world of apps and	852
799	and Christopher Potts. 2024. DSPy: Compiling	people for benchmarking interactive coding agents.	853
800	declarative language model calls into self-improving	In <i>Proceedings of the 62nd Annual Meeting of the</i>	854
801	pipelines. In <i>International Conference on Learning</i>	<i>Association for Computational Linguistics</i> .	855
802	<i>Representations</i> .		
803	Jiazheng Li, Emine Yilmaz, Bei Chen, and Dieu-	Rakshith Vasudev, Melisa Russak, Dan Bikel, and	856
804	Thu Le. 2026. Towards self-improving error di-	Waseem Alshikh. 2026. Accurate failure prediction	857
805	agnosis in multi-agent systems. <i>arXiv preprint</i>	in agents does not imply effective failure prevention.	858
806	<i>arXiv:2604.17658</i> .	<i>arXiv preprint arXiv:2602.03338</i> .	859
807	Xiaofeng Lin, Yingxu Wang, Tung Sum Thomas Kwok,	Abraham Wald. 1945. Sequential tests of statistical	860
808	Daniel Guo, Sahil Arun Nale, Charles Fleming,	hypotheses. <i>The Annals of Mathematical Statistics</i> ,	861
809	and Guang Cheng. 2026. REFLECT: Intervention-	16(2):117–186.	862
810	supported error attribution for silent failures in LLM		
811	agent traces. <i>arXiv preprint arXiv:2606.09071</i> .	Mengting Wan, Tara Safavi, Sujay Kumar Jauhar, Yujin	863
		Kim, Scott Counts, Jennifer Neville, Siddharth Suri,	864
812	Jiale Liu, Huajun Xi, Shaokun Zhang, Yifan Zeng, Tian-	Chirag Shah, Ryen W. White, Longqi Yang, Reid	865
813	wei Yue, Chi Wang, Jian Kang, Qingyun Wu, and	Andersen, Georg Buscher, Dhruv Joshi, and Nagu	866
814	Huazheng Wang. 2026. Who&when pro: Can LLMs	Rangan. 2024. TnT-LLM: Text mining at scale with	867
815	really attribute failures in AI agents? <i>arXiv preprint</i>	large language models. In <i>Proceedings of the 30th</i>	868
816	<i>arXiv:2607.09996</i> .	<i>ACM SIGKDD Conference on Knowledge Discovery</i>	869
		<i>and Data Mining</i> .	870
817	Ming Ma, Jue Zhang, Fangkai Yang, Yu Kang, Qing-	Xiaojun Wu, Cehao Yang, Honghao Liu, Xueyuan Lin,	871
818	wei Lin, Saravan Rajmohan, and Dongmei Zhang.	Wenjie Zhang, Zhichao Shi, Xuhui Jiang, Chengjin	872
819	2025. DoVer: Intervention-driven auto debug-	Xu, Jia Li, and Jian Guo. 2026. Bayesian-Agent:	873
820	ging for LLM multi-agent systems. <i>arXiv preprint</i>	Posterior-guided skill evolution across LLM agent	874
821	<i>arXiv:2512.06749</i> .	harnesses. <i>arXiv preprint arXiv:2606.08348</i> .	875
		Peng Xia, Rujun Han, Zifeng Wang, Yanfei Chen, Yu-	876
		fan Zhuang, Yoonho Lee, Chengsong Huang, Han	877

Yu, Zhongying CuiZhu, Yifei Ming, Huaxiu Yao, Burak Gokturk, Tomas Pfister, and Chen-Yu Lee. 2026. RRSI: Regularized recursive self-improvement of agent harnesses. *arXiv preprint arXiv:2609.24972*.

Chengrun Yang, Xuezhi Wang, Yifeng Lu, Hanxiao Liu, Quoc V. Le, Denny Zhou, and Xinyun Chen. 2024. Large language models as optimizers. In *International Conference on Learning Representations*.

Shunyu Yao, Noah Shinn, Pedram Razavi, and Karthik Narasimhan. 2024. τ -bench: A benchmark for tool-agent-user interaction in real-world domains. *arXiv preprint arXiv:2406.12045*.

Mert Yuksekgonul, Federico Bianchi, Joseph Boen, Sheng Liu, Pan Lu, Zhi Huang, Carlos Guestrin, and James Zou. 2025. Optimizing generative AI by back-propagating language model feedback. *Nature*, 639.

Guibin Zhang, Junhao Wang, Junjie Chen, Wangchunshu Zhou, Kun Wang, and Shuicheng Yan. 2026a. AgenTracer: Who is inducing failure in the LLM agentic systems? In *International Conference on Learning Representations*. ArXiv:2509.03312.

Jenny Zhang, Shengran Hu, Cong Lu, Robert Lange, and Jeff Clune. 2025a. Darwin Gödel machine: Open-ended evolution of self-improving agents. *arXiv preprint arXiv:2505.22954*.

Qizheng Zhang, Changran Hu, Shubhangi Upasani, Boyuan Ma, Fenglu Hong, Vamsidhar Kamanuru, Jay Rainton, Chen Wu, Mengmeng Ji, Hanchen Li, Urmish Thakker, James Zou, and Kunle Olukotun. 2026b. Agentic context engineering: Evolving contexts for self-improving language models. In *International Conference on Learning Representations*.

Shaokun Zhang, Ming Yin, Jieyu Zhang, Jiale Liu, Zhiguang Han, Jingyang Zhang, Beibin Li, Chi Wang, Huazheng Wang, Yiran Chen, and Qingyun Wu. 2025b. Which agent causes task failures and when? on automated failure attribution of LLM multi-agent systems. In *Proceedings of the 42nd International Conference on Machine Learning*. ArXiv:2505.00212.

Andrew Zhao, Daniel Huang, Quentin Xu, Matthieu Lin, Yong-Jin Liu, and Gao Huang. 2024. ExpeL: LLM agents are experiential learners. In *Proceedings of the AAAI Conference on Artificial Intelligence*.

Kunlun Zhu, Zijia Liu, Bingxuan Li, Muxin Tian, Yingxuan Yang, Jiaxun Zhang, Pengrui Han, Qipeng Xie, Fuyang Cui, Weijia Zhang, Xiaoteng Ma, Xiaodong Yu, Gowtham Ramesh, Jialian Wu, Zicheng Liu, Pan Lu, James Zou, and Jiaxuan You. 2025. Where LLM agents fail and how they can learn from failures. *arXiv preprint arXiv:2509.25370*.

Kunlun Zhu, Xuyan Ye, Zhiguang Han, Yuchen Zhao, Bingxuan Li, Weijia Zhang, Muxin Tian, Xiangru Tang, Pan Lu, James Zou, Jiaxuan You, and Heng Ji. 2026. AgentDebugX: An open-source toolkit for failure observability, attribution, and recovery in LLM agents. *arXiv preprint arXiv:2607.18754*.

A An Accounting Framework for Single-Round Repair

Setting. An agent is assembled from *components* j (prompt sections, tool descriptions, configuration). On a task distribution it fails with rate F ; an LLM groups its failures into categories i with shares f_i . For each *cell* (i, j) that the analyst blames, it proposes K candidate patches k that rewrite component j . Let $e_{ij} \in \{0, 1\}$ indicate whether changing j can repair category i at all, and q_k the fraction of category- i failures that patch k repairs if so. Separating e from q matters because a patch that fails to help may be aimed at the wrong component or merely be a bad patch. We call a cell with $e_{ij} = 1$ an *effective edge*. This is an *interventional* notion: some change to j repairs failures of category i . It need not mean that j was the original cause of those failures.

Net effect. Applying patch k of cell (i, j) changes the success rate by

$$\Delta_k = F f_i e_{ij} q_k - (1 - F) \rho_k - c_a - c_w (1 - e_{ij}), \quad (2)$$

where ρ_k is the patch’s absolute excess failure rate on tasks that used to succeed (regressions, which may hit any category): such a task fails a re-run with probability r_0 unpatched and $r_0 + \rho_k$ patched, so $\rho_k < 0$ is possible; $c_a \geq 0$ is a cost charged for every applied change (review, maintenance) and $c_w \geq 0$ a penalty charged only for a change aimed at a wrong component (risk not captured by ρ_k). Costs are fractions of the failure mass F . The goal is to choose a set of patches, at most one per cell, that maximises $\sum_k \Delta_k$ on held-out tasks; effects of patches to different components are treated as additive, and held-out success of whole patch sets (Table 1) and a leave-one-out ablation (Appendix K) are the only checks. A targeted replay of a category- i failure with patch k succeeds with probability $b_i + (1 - b_i) e_{ij} q_k$, a null replay with probability b_i , and a regression run fails with probability $r_0 + \rho_k$.

Value of verification. Before any evidence, the net effects $\Delta = (\Delta_k)$ of the candidate patches are uncertain, with a prior that reflects the analyst’s accuracy and what is known about patch quality and regressions. The *default* applies, in each cell, the patch with the largest $\mathbb{E}[\Delta_k]$ if that is positive. Verification observes outcomes Y and then chooses; its value is $V(Y) = \mathbb{E}[\max_S \mathbb{E}[\sum_{k \in S} \Delta_k \mid Y]] - \max_S \mathbb{E}[\sum_{k \in S} \Delta_k]$, before the cost of the

experiments. Write $x^+ = \max(x, 0)$ and $x^- = \max(-x, 0)$.

Proposition 1 (Value of verification). *For any experiments Y , $V(Y)$ is at most the value of learning Δ exactly, which is the sum over cells of*

$$V_{ij} = \mathbb{E}[\max_k \Delta_k^+] - \max_k (\mathbb{E}\Delta_k)^+. \quad (3)$$

With k^* the default patch of the cell,

$$V_{ij} \leq \underbrace{\mathbb{E}[\max_k \Delta_k^+ - \Delta_{k^*}^+]}_{\text{choosing among patches}} + \underbrace{\mathbb{E}[\Delta_{k^*}^-]}_{\text{avoiding harm}}, \quad (4)$$

with equality when the default applies k^* , and $\mathbb{E}[\Delta_{k^*}^-] \leq (1 - F) \mathbb{E}[\rho_{k^*}^+] + c_a + c_w P(e_{ij}=0)$.

Corollary 1 (Edge-only verification). *Suppose a cell has a single candidate patch whose net effect depends only on e_{ij} : $\Delta = G$ if $e_{ij} = 1$ and $\Delta = -H$ if $e_{ij} = 0$, with $G, H \geq 0$ known. With $\pi = P(e_{ij}=1)$, $V_{ij} = \min(\pi G, (1 - \pi)H)$.*

What survives deployment constraints. Deployment may use a heuristic default D , applies at most one patch per component, which couples cells, combines repairs of one category as $1 - \prod_k (1 - q_k)$ (overstated by at most $f_i \sum_{k < l} q_k q_l$ when added), and cannot push the failure rate on solved tasks below zero ($-\sum_k \rho_k \leq r_0$). Three statements survive. (i) The first sentence of the proposition holds for any set of admissible actions, so $V(Y)$ is at most the value of perfect information over deployable sets. (ii) Over a heuristic default that value is $V + G_D$, where $G_D = \max_S \mathbb{E}[\sum_{k \in S} \Delta_k] - \mathbb{E}[\sum_{k \in D} \Delta_k] \geq 0$ is the decision-rule gap that better use of the prior alone would close. (iii) The value of dropping harmful default patches is at most their expected harm, $\sum_{k \in D} \mathbb{E}[(1 - F)\rho_k^+ + c_a + c_w(1 - e_{ij})]$, because dropping a patch from an overlapping set loses at most its marginal repair. What does not survive is the separation over cells: with one patch per component, the choice and discovery terms have no closed form, and we compute them in sequence (harm, then choice on the default's components, then any component) with a greedy optimiser, which gives a lower bound on the constrained value of perfect information under a given harm model. The three sources of value are *avoiding harm*, which needs regressions or change costs because replay success alone cannot create it; *choosing among patches* of different quality; and *discovery* of edges the analyst missed, the whole term $\mathbb{E}[\max_k \Delta_k^+]$ in cells

	Default rule	Value	Gap	Harm	Choice	Discov.	Total
Retail	Apply, $c = 0$	1.43	0.06	0.41	0.57	0.20	1.18
	Bayes, $c = 0$	1.49	–	0.80	0.32	0.00	1.12
	Apply, $c_w = 0.02$	1.10	–0.19	0.54	0.56	0.03	1.13
	normal, partial pooling	0.77		0.87	0.98	0.17	2.02
	normal, $\tau=0$	1.12		0.23	0.22	0.42	0.87
	no pooling	–1.82		3.30	1.21	0.13	4.65
	tasks resampled per cell	–0.52		2.28	0.90	0.13	3.31
Airline	Apply, $c = 0$	0.66	–0.13	0.20	0.36	0.29	0.86
	Bayes, $c = 0$	0.53	–	0.55	0.44	0.00	0.98
	Apply, $c_w = 0.02$	0.54	0.02	0.27	0.36	0.11	0.74
	normal, partial pooling	0.73		0.04	0.25	0.40	0.69
	normal, $\tau=0$	0.73		0.04	0.25	0.40	0.69
	no pooling	0.39		0.91	0.10	0.15	1.15
	tasks resampled per cell	0.63		0.39	0.25	0.23	0.87
		0.86		0.15	0.11	0.30	0.56

Table 3: Value of perfect verification over two no-evidence defaults, deployable sets, posterior means under the task-level binomial harm model (Appendix J) as fractions of the repair-only oracle. *Gap*: Bayes default minus Apply attributed (G_D with the sign reversed; negative when the heuristic does better). Harm, choice and discovery are added in that order: dropping default patches with negative net effect, then the best cell and patch on the default's components, then any other component. Indented rows give the value over Apply at $c = 0$ under the normal-approximation harm models of an earlier analysis (partial, complete and no pooling, tasks resampled before fitting) and with the additive per-cell form of Proposition 1 instead of deployable sets. A value of 1 is 0.070 (retail) or 0.114 (airline) in absolute success.

the default leaves alone. The corollary fails when quality or regressions are uncertain given the edge ($\pi = 1$ with $\Delta = +0.10$ or -0.05 equally likely gives $V = 0.025$).

Proposition 2 (Cost of evidence). *Under Wald's approximations, a sequential test that decides e_{ij} from targeted replays with error δ needs on average about $\log \frac{1-\delta}{\delta} / \text{KL}(b + (1-b)q \parallel b)$ replays when the edge is real. Detecting a regression of size ρ against a base rate r_0 with a one-sided level- α test and power $1 - \beta$ needs about $(z_\alpha + z_\beta)^2 r_0 (1 - r_0) / \rho^2$ regression runs.*

Two consequences follow. First, targeted replays are uninformative about the harm term, because they only see the target category: a patch can pass every replay and still break other tasks. Second, small regressions are expensive to see. With $r_0 = 0.1$, a regression of $\rho = 0.25$ is detected with 90% power by 20 runs, whereas $\rho = 0.03$ needs about 850. Proposition 1 is the standard value-of-information argument (Chaloner and Verdinelli, 1995); what is new is its decomposition for patches and its evaluation on measured banks.

Where could verification add value? Table 3 evaluates Proposition 1 on deployable sets. Every value in it is conditional on a model of the harm ρ_k that the regression runs cannot pin down per patch, so we read only its qualitative pattern. *Defaults.* The Bayes default, built from the net-effect method’s priors, scores 1.49 oracles on retail against 1.43 for applying the attributed patches, and 0.53 against 0.66 on airline; $G_D \geq 0$ holds only under the prior, so the negative realised gap on airline means the edge prior is too generous there. *Value.* Under the binomial model, perfect information is worth 1.18 oracles over the heuristic on retail (90% interval [0.10, 2.99], about 8 points of absolute success) and 0.86 on airline ([0.24, 1.55], about 10 points), split into 0.41, 0.57 and 0.20 (retail) and 0.20, 0.36 and 0.29 (airline) for harm, choice and discovery. Under the earlier normal models the retail value ranges from 0.87 (complete pooling) to 4.65 (no pooling) and airline’s from 0.69 to 1.15; dropping the support constraint, so that the uncertain benefits of every unblamed patch add up without limit, gives 9.5 on retail (Table 14). Only coarse features are stable: all three terms contribute, and most of the value needs evidence about patches the default applies (harm and choice) rather than discovery of missed edges. Proposition 1 names the quantities that could make verification pay, but it cannot decide whether to verify a new diagnosis: its most influential input, each patch’s harm, takes hundreds of regression runs to measure (Proposition 2), and its values move several-fold with the harm model.

B Study 1: Measurements and Net Value

Tables 4 and 5 give the measurements and the $\text{Net}(N)$ values discussed in §4.

C Proofs

Proposition 1. For any signal Y and any action set, $\max_S \mathbb{E}[U(S) \mid Y] \leq \mathbb{E}[\max_S U(S) \mid Y]$, so taking expectations gives $V(Y) \leq \mathbb{E}[\max_S U(S)] - \max_S \mathbb{E}[U(S)]$, the value of perfect information; this is the standard argument and needs no assumption on how Y relates to Δ . With $U(S) = \sum_{k \in S} \Delta_k$ and at most one patch per cell, both maxima separate over cells: $\max_S U(S) = \sum_{ij} \max_k \Delta_k^+$ and $\max_S \mathbb{E}[U(S)] = \sum_{ij} \max_k (\mathbb{E}\Delta_k)^+$, which gives V_{ij} ; no independence between cells is needed. For Eq. 4, let k^* maximise $\mathbb{E}\Delta_k$. If $\mathbb{E}\Delta_{k^*} \geq 0$,

	ShopDesk	Retail	Airline
Analyst accuracy (audited)	0.72	0.41	0.29
Effective edges among blamed cells	–	0.18	0.33
Judge check: cost / replay	0.72	0.28	0.23
sens. / FPR, vs. single replay	0.57 / 0.21	0.37 / 0.11	0.41 / 0.13
sens. / FPR, vs. patch replay rate	–	0.49 / 0.05	0.55 / 0.07
efficiency ratio vs. replays	–	0.88 [0.46, 1.53]	1.25 [0.19, 3.95]
Spurious recovery b (range)	–	0.23 (0–0.50)	0.20 (0–0.50)
Repair q of effective edges (median)	–	0.50	0.41
Replays to decide an edge	–	5	7
Unpatched failure on solved tasks	–	0.15	0.12
Mean harm $\bar{\rho}$ (points)	–	–1.0 [–3.4, 1.5]	–0.6 [–3.3, 1.9]
Spread of ρ_k (points)	–	2.6 [0.2, 5.3]	2.1 [0.2, 4.5]
Patches harmful / helpful (95%)	–	2 / 5	3 / 4

Table 4: What the evidence costs and shows. The efficiency ratio is the judge’s break-even cost ratio divided by its actual cost (Proposition 4); values above 1 favour judge checks (Appendix I). Harm rows: excess failure rate of a patch on previously solved tasks under the task-level binomial model (Appendix J), posterior mean and 90% interval; the last row counts patches whose 20 regression runs differ significantly from the task-matched baseline.

Verified set	Δp [90%]	Δc_{run}	C	Net(1000) [90%], episodes	
				$v = 1$	$v = 10$
<i>Retail</i>					
Replay-each	−4.0 [−9.8, 1.8]	−0.06	10	12 [−217, 242]	−348 [−966, 271]
HarnessFix-style, per patch	−4.3 [−12.8, 4.2]	−0.03	9	−18 [−204, 167]	−408 [−1273, 456]
HarnessFix-style, bundle	−2.7 [−5.0, −0.4]	−0.00	6	−31 [−244, 183]	−271 [−583, 42]
Uncertainty sampling	−1.7 [−6.9, 3.5]	−0.07	10	41 [−189, 271]	−109 [−676, 458]
Uncertainty + checks	−2.8 [−8.4, 2.7]	0.02	10	−56 [−285, 173]	−311 [−907, 286]
CARVE, replay only	0.3 [−5.3, 6.0]	−0.02	10	10 [−220, 240]	40 [−568, 648]
CARVE	−3.7 [−9.9, 2.6]	−0.04	10	−8 [−227, 210]	−338 [−999, 322]
CARVE, measured check	−2.7 [−7.6, 2.3]	−0.01	10	−24 [−254, 206]	−264 [−810, 282]
CARVE, robust check	−1.2 [−6.3, 4.0]	0.02	9	−37 [−263, 188]	−142 [−704, 420]
Net-effect	−4.3 [−9.9, 1.3]	0.05	10	−100 [−307, 106]	−490 [−1083, 103]
<i>Airline</i>					
HarnessFix-style, per patch	−4.0 [−11.9, 3.9]	−0.04	8	−7 [−226, 212]	−367 [−1185, 451]
HarnessFix-style, bundle	−3.3 [−9.3, 2.6]	−0.06	6	23 [−209, 254]	−277 [−911, 356]
CARVE	−0.7 [−8.8, 7.5]	−0.11	10	96 [−156, 349]	36 [−814, 887]
Net-effect	−0.3 [−9.1, 8.5]	−0.08	10	68 [−194, 330]	38 [−875, 951]

Table 5: $\text{Net}(N)$ of each verified set against Apply attributed (Eq. 1), at $N = 1000$ deployed episodes, with no change cost and a success worth v episodes of running cost. Δp : held-out difference in points (Table 1); Δc_{run} : estimated change in running cost per deployed episode, from the tokens of patched agents on solved tasks and of truncated step-limit episodes; C : verification episodes. Intervals combine the uncertainty of Δp and of Δc_{run} (standard error about 0.1), as normal.

then $V_{ij} = \mathbb{E}[\max_k \Delta_k^+] - \mathbb{E}[\Delta_{k^*}^+] + \mathbb{E}[\Delta_{k^*}^-]$, which is Eq. 4 with equality. Otherwise $V_{ij} = \mathbb{E}[\max_k \Delta_k^+] = \mathbb{E}[\max_k \Delta_k^+ - \Delta_{k^*}^+] + \mathbb{E}[\Delta_{k^*}^+]$, and $\mathbb{E}[\Delta_{k^*}^+] < \mathbb{E}[\Delta_{k^*}^-]$ because $\mathbb{E}\Delta_{k^*} < 0$. Finally, the repair term $F f_{ij} e_{ij} q_k$ is non-negative, so $\Delta_k^- \leq (1 - F)\rho_k^+ + c_a + c_w(1 - e_{ij})$; take expectations.

Heuristic defaults and constraints. For a default D , $\mathbb{E}[\max_S U(S)] - \mathbb{E}[U(D)] = (\mathbb{E}[\max_S U(S)] - \max_S \mathbb{E}[U(S)]) + (\max_S \mathbb{E}[U(S)] - \mathbb{E}[U(D)])$, and the second bracket is $G_D \geq 0$ by definition of the maximum. Restricting S to deployable sets leaves the argument unchanged. For repairs in

one category, the Bonferroni inequality gives $1 - \prod_k (1 - q_k) \geq \sum_k q_k - \sum_{k < l} q_k q_l$, which bounds the overstatement of the additive form. The bound on $-\sum_k \rho_k$ holds because the patched failure rate $r_0 + \sum_k \rho_k$ on solved tasks is non-negative under additivity. For (iii), removing patch k from a set S changes its value by $-\text{rep}(S) + \text{rep}(S \setminus k) + (1 - F)[\bar{\rho}(S) - \bar{\rho}(S \setminus k)] + c_a + c_w(1 - e_{ij})$, where rep is the product-form repair and $\bar{\rho}$ the capped sum of harms. Repair is monotone in S , so the first bracket is non-negative, and capping $\sum \rho$ from below changes it by at most ρ_k^+ when k is removed; summing over the dropped patches and taking expectations gives the bound. The greedy optimiser keeps, per component, the cell and patch with the largest positive net effect, which is a feasible deployable set, so the value it attains is a lower bound on the constrained optimum.

Corollary 1. With one patch, $V = \mathbb{E}[\Delta^+] - (\mathbb{E}\Delta)^+ = \pi G - (\pi G - (1 - \pi)H)^+ = \min(\pi G, (1 - \pi)H)$. The assumption that Δ is a function of e_{ij} is what makes the edge state sufficient: if q_k or ρ_k is uncertain given e_{ij} , learning only e_{ij} leaves value on the table and the general expression applies.

Proposition 2. The first part is Wald’s approximation for a sequential probability ratio test between $p_1 = b + (1 - b)q$ and $p_0 = b$: with thresholds $\pm \log \frac{1-\delta}{\delta}$, neglecting overshoot, the expected sample size under p_1 is the threshold divided by the per-observation drift $\text{KL}(p_1 \| p_0)$. The second is the normal-approximation sample size of a one-sided test of a binomial proportion, evaluated with the variance at r_0 .

D Verification Policies on the Intervention Banks

Summary. The banks give the same answer as the deployment comparison without new runs. In that comparison the policy seed added no detectable variance: the variance of the seed means of the verified sets’ differences from Apply attributed is 0.0007, against 0.0019 expected from run-to-run noise alone. Under the task-level harm model, applying the attributed patches is worth 1.43 (retail) and 0.66 (airline) of the repair-only oracle, and the verifying policies 0.39–1.56 at budget 40, beating it in 19–57% of posterior draws; scored on bank halves that share no trajectory with the policies’ evidence, they beat it in 22–47%. Policies that use

the analyst’s blame as an edge prior used an analyst accuracy of 0.7; with the audited 0.41 and 0.29, or half of these, every such policy is worth less than with 0.7, on both domains (Table 8).

Policies. *Net-effect allocation.* A spike-and-slab model per cell takes its edge prior from the attribution counts (Eq. 7, Appendix S), with grid posteriors for q_k and Beta posteriors for each b_i and each patch’s regression failure rate r_k . It estimates Eq. 2 as $P(e_{ij}=1 \mid \mathcal{D}) F f_i \mathbb{E}[q_k \mid \mathcal{D}] - (1 - F) \max(0, \mathbb{E}[r_k \mid \mathcal{D}] - r_0)/(1 - r_0) - c$, a stricter objective than Eq. 2: it charges c on every applied change, never credits a patch for lowering failures on solved tasks, and scales harm by $1/(1 - r_0)$; Table 12 ablates the last two. It repeatedly runs the targeted replay, null replay or regression run with the largest n -step knowledge gradient per unit cost, $n \in \{1, 2, 4, 8\}$ (Frazier et al., 2008), and applies every patch with positive posterior net effect. *Apply attributed* uses no evidence: for each category it applies the first patch of the most-blamed component and of any component with at least 40% of the blame. *Replay-each* tests hypotheses in order of blame with three replays and accepts on two successes, as in do-then-verify methods (Ma et al., 2025). *HarnessFix-style, per patch* adds two regression runs that must break nothing; these counts are ours. *HarnessFix-style, bundle gate* is closer to the original, which validates a whole bundle of repairs on held-out tasks and promotes it if it solves at least one more task and breaks at most two (Chen et al., 2026); we apply the *Apply attributed* bundle, draw validation episodes from the bank, and allow three attempts. *Uncertainty sampling* replays the cell with the largest frequency-weighted posterior uncertainty, with or without three cheap checks per patch first. CARVE and its variants are described in Appendix S.

Scoring. Policies run offline by drawing recorded outcomes from the bank, with replacement (without, as a check), with costs in measured tokens and a full replay as the unit; resampling seeds measure policy variability given the bank, not new evidence. *Adjudication*: a cell is an effective edge if its best patch’s replay success exceeds the category’s null rate by at least 0.2 (one-sided Fisher $p < 0.05$), giving 4 effective edges on retail and 7 on airline; choosing the best of three patches on the same replays makes this label optimistic, so we also score policies on bank halves they did not see. *Harm*: every patch’s ρ_k is drawn from the task-level bino-

mial model of Appendix J, and edges and repair rates from a bootstrap of the replays. *Deployable sets*: every accepted set is reduced to one patch per component, keeping the cell of the most frequent category, as on held-out tasks. Values are fractions of the *repair-only oracle*, the failure mass repaired by the effective edges’ best patches with no cost, a fixed yardstick worth 0.070 (retail) and 0.114 (airline) in absolute success. The unpatched baseline (10 runs on each of 111 regression tasks) was collected about 12 hours after the regression runs; unpatched held-out success did not change over that time (0.816 vs. 0.817 on clean retail episodes). *Protocol*: retail was the development domain for every method, including the variants in Appendix L; airline was evaluated once for the original policies and once, later, for the net-effect method, which was frozen on retail beforehand.

Results. Table 6 reports each policy’s posterior expected value. On retail, applying the attributed patches is worth 1.43 oracles (10.1 points of success) and is the best policy at budget 10; at 40 and 80, CARVE with a robust check is ahead (1.56 and 1.59) but beats applying everything in only 57% of draws at budget 40; across budgets, the other verifying policies are worth 0.55–1.49. On airline, applying everything scores 0.66 and no verifying policy exceeds 0.73. The 90% intervals span about three oracles on retail (applying everything: $[-0.52, 2.67]$) and one and a half on airline. On source-split halves every value falls, because each half supports fewer effective edges: applying everything scores 0.96 (retail) and 0.23 (airline), and verifying policies beat it in 22–42% (retail) and 35–47% (airline) of draws at budget 40. With $c_w = 0.02$ per wrong change (Table 7) applying everything falls to 1.10 and 0.54 on the full banks; the leaders change with budget and split, and none leads everywhere. On the banks, as on held-out tasks, no verification policy reliably beats applying the attributed patches.

Priors. Every policy with an edge prior used an analyst accuracy of $\hat{r} = 0.7$, well above the audited 0.41 (retail) and 0.29 (airline). Table 8 reruns their budget-40 decisions with the audited accuracy and with half of it. Every such policy is worth less than with 0.7 on both domains, by 0.11 to 0.65 oracles on retail and 0.04 to 0.25 on airline at the audited accuracy, and halving it again changes values by at most 0.30 more. With the audited or the weaker accuracy, no verifying policy is worth more than

applying the attributed patches.

E Significance-Thresholded Scoring

Charging only regressions significant against the paired-trial r_0 reproduces the picture of a scoring that sets uncertain harm to zero: applying everything recovers 84% and 60% of the oracle and no verifying policy reaches it. With $c_w = 0.02$ per wrong change, the per-patch HarnessFix-style rule (retail, budgets 10–40; airline, 40) and the net-effect method (retail, 80; airline, 80, tied with uncertainty sampling with checks) lead, and CARVE declines with budget on airline (0.30→0.19). Appendix G varies c for both cost types; the winner changes with c .

F Robustness at Budget 40

G Change-Cost Sweep

Table 11 varies the change cost. The best policy changes with c : applying the attributed patches is best on retail at $c = 0$ (on airline, CARVE told $c = 0$ edges it, 0.71 against 0.60), replay-each at $c = 0.01$ on retail, the per-patch HarnessFix-style rule from $c = 0.01$ to 0.02 on airline and at 0.02 on retail, and the net-effect method at $c = 0.05$ on retail under both cost types and on airline per applied change. Charging every applied change rather than only wrong ones lowers every policy but changes the winner in only one setting (airline, $c = 0.05$). Told $c = 0$, the policies that use a cost internally accept patches with significant regressions, which is why some of them are negative in the first column.

H Policy Inputs and Regression Sampling

Attribution, taxonomy induction, labelling and candidate patches are shared by all policies and not charged to any budget. Every policy that uses an edge prior uses Eq. 7 with analyst accuracy $\hat{r} = 0.7$, not the audited 0.41 and 0.29 of Table 4; Table 8 rescores them with the audited accuracy and half of it. Three further inputs come from the bank itself and are also uncharged: the cost of each kind of experiment (mean tokens), the mean spurious recovery $\hat{b}$, which CARVE and uncertainty sampling plug in and the net-effect method uses only as a prior mean for its per-category b_i , and, for the two CARVE variants that use them, the judge’s measured sensitivity and false-positive rate. The re-run failure rate r_0 comes from paired re-runs of training

Budget (episodes)	τ^2 retail (development)						τ^2 airline					
	full bank			source-split halves			full bank			source-split halves		
	10	40	80	10	40	80	10	40	80	10	40	80
Apply attributed	1.43	1.43	1.43	0.96	0.96	0.96	0.66	0.66	0.66	0.23	0.23	0.23
Bayes default (Prop. 1)	1.49	1.49	1.49	—	—	—	0.53	0.53	0.53	—	—	—
Replay-each	1.37	1.26	1.35	0.85	0.81	0.83	0.60	0.53	0.54	0.18	0.15	0.11
HarnessFix-style, per patch	0.55	1.01	1.28	0.55	0.87	0.97	0.19	0.45	0.53	0.09	0.24	0.22
HarnessFix-style, bundle	0.68	1.16	1.31	0.38	0.56	0.67	0.36	0.57	0.60	0.12	0.17	0.17
Uncertainty sampling	1.27	1.39	1.32	0.84	0.94	0.94	0.59	0.65	0.67	0.10	0.22	0.24
Uncertainty + checks	1.36	1.33	1.35	0.73	0.87	0.98	0.54	0.58	0.65	0.04	0.14	0.18
CARVE, replay only	1.23	1.41	1.49	0.87	0.91	0.98	0.71	0.65	0.70	0.24	0.24	0.24
CARVE	1.30	1.12	0.93	0.61	0.54	0.63	0.63	0.39	0.25	0.11	0.08	0.09
CARVE, measured check	1.23	1.41	1.49	0.82	0.93	1.01	0.71	0.65	0.71	0.24	0.23	0.23
CARVE, robust check	1.35	1.56	1.59	0.84	0.96	1.02	0.68	0.64	0.73	0.26	0.24	0.23
Net-effect (ours)	1.02	1.09	1.29	0.82	0.87	0.97	0.54	0.60	0.64	0.14	0.18	0.19

Table 6: Posterior expected value of each policy’s deployable patch set on the τ^2 banks at $c = 0$, under the task-level binomial harm model, as a fraction of the repair-only oracle; bold is the best per column. *Full bank*: policies and scoring use the same bank; 300 posterior draws, each also bootstrapping the replays, averaged over 40 policy seeds. *Source-split halves*: the bank is split into halves that share no source trajectory and no regression task; policies run on one half, with their inputs (experiment costs, spurious recovery, judge calibration) from that half and the audited analyst accuracy, and are scored on the other (5 halvings $\times$ 100 draws, 20 seeds). The Bayes default is Proposition 1’s no-evidence rule under the net-effect method’s priors. Values above 1 arise when patches are estimated to reduce failures on previously solved tasks.

tasks. *Regression tasks* are drawn uniformly, with replacement, from the clean training episodes the agent solved, so they follow the training-task distribution of the same domain, not the held-out tasks, which also include fault-injected episodes. The extension to 20 runs covered every patch an original policy accepted; 98–100% of the patches that any policy, including the later net-effect method, deploys at budget 40 have 20 runs, and a policy asking for more runs of a 4-run patch resamples those 4.

I Judge Configurations

The three new configurations regenerate the failing step once per pair and judge it with each prompt; the short prompt shows the task, the diagnosed root cause, the tool results or user message just before the failing step, and the old and new actions. Pairs are stratified on the paired replay’s outcome; because a check and its paired replay are separate regenerations, the check is independent of that outcome given the patch, so the stratification does not bias the patch-level fit. The short-prompt judges agree with the bank judge on 64–76% of pairs. The experiment used about 310 episode-equivalents of tokens. *Why the two definitions differ*. Conditioning on a single replay’s outcome attenuates both quantities towards the judge’s overall pass rate, because many failed replays come from patches that

do repair some failures and many successful ones from spurious recovery; the patch-level fit removes this noise, which raises the bank judge’s break-even ratio from about 0.1 to 0.25 (retail) and 0.29 (airline). Across categories, using each category’s own spurious recovery and pairs (≥ 20), the single-replay ratio of the bank judge ranges from 0 to 2.0 on retail and 0.04 to 1.5 on airline. *Latency* was not measured; counted in sequential model calls, a check is two (regeneration and judge) and a replay from the failing step a median of 6 (retail) or 7 (airline), interquartile range 4–11. *Money*: we cannot state prices for the models we used that will remain accurate, so the table reports token cost with the analyst model priced at one, two and four times the agent model.

J Harm Model, Checks and Sensitivity

Model. Regression outcomes are modelled per patch and task, together with the unpatched baseline runs on the same tasks. With n_{0t} unpatched runs of regression task t and f_{0t} failures, and n_{kt} runs of patch k on task t with f_{kt} failures,

$$f_{0t} \sim \text{Bin}(n_{0t}, \sigma(\alpha_t)),$$

$$f_{kt} \sim \text{Bin}(n_{kt}, \sigma(\alpha_t + \beta(\alpha_t - \bar{\alpha}) + \delta_k)),$$

with $\alpha_t \sim N(a, s_a^2)$, $\delta_k \sim N(\mu, s_d^2)$, $a, \mu \sim N(0, 3^2)$, $s_a \sim \text{HalfNormal}(2)$, $s_d \sim \text{HalfNormal}(1)$, $\beta \sim N(0, 1)$, and $\bar{\alpha}$ the logit of

Budget (episodes)	τ^2 retail (development)						τ^2 airline					
	full bank			source-split halves			full bank			source-split halves		
	10	40	80	10	40	80	10	40	80	10	40	80
Apply attributed	1.10	1.10	1.10	0.61	0.61	0.61	0.54	0.54	0.54	0.06	0.06	0.06
Bayes default (Prop. 1)	0.91	0.91	0.91	–	–	–	0.56	0.56	0.56	–	–	–
Replay-each	1.10	1.03	1.13	0.54	0.52	0.51	0.49	0.45	0.44	0.02	0.01	–0.06
HarnessFix-style, per patch	0.55	0.93	1.10	0.55	0.80	0.78	0.18	0.42	0.48	0.06	0.18	0.10
HarnessFix-style, bundle	0.52	0.89	1.02	0.22	0.30	0.40	0.29	0.46	0.48	0.02	0.01	0.02
Uncertainty sampling	1.00	1.20	1.13	0.79	0.82	0.74	0.53	0.59	0.61	0.03	0.13	0.12
Uncertainty + checks	1.10	1.19	1.21	0.69	0.79	0.85	0.47	0.53	0.60	–0.02	0.06	0.07
CARVE, replay only	0.92	1.14	1.15	0.75	0.71	0.67	0.62	0.56	0.60	0.13	0.11	0.07
CARVE	1.08	1.02	0.89	0.56	0.44	0.49	0.55	0.33	0.21	0.04	–0.02	–0.03
CARVE, measured check	0.92	1.14	1.15	0.70	0.73	0.69	0.63	0.56	0.62	0.14	0.10	0.06
CARVE, robust check	1.04	1.27	1.25	0.71	0.75	0.70	0.61	0.56	0.64	0.16	0.11	0.06
Net-effect (ours)	0.87	0.98	1.15	0.77	0.78	0.83	0.49	0.55	0.58	0.08	0.08	0.06

Table 7: As Table 6, with a cost of $c_w = 0.02$ of the failure mass charged per patch aimed at a wrong edge.

	Retail			Airline		
	0.7	0.41	0.20	0.7	0.29	0.15
Analyst accuracy $\hat{r}$	0.7	0.41	0.20	0.7	0.29	0.15
Apply attributed	1.43	1.43	1.43	0.66	0.66	0.66
Uncertainty sampling	1.39	1.02	0.95	0.65	0.40	0.35
Uncertainty + checks	1.33	0.99	1.00	0.58	0.35	0.26
CARVE, replay only	1.41	1.22	1.10	0.65	0.57	0.49
CARVE	1.12	0.47	0.17	0.39	0.17	0.12
CARVE, measured check	1.41	1.22	1.10	0.65	0.61	0.50
CARVE, robust check	1.56	1.27	1.14	0.64	0.53	0.47
Net-effect	1.09	0.98	0.99	0.60	0.42	0.40

Table 8: Policies that use the edge prior of Eq. 7, with the analyst accuracy used in the main analysis (0.7), the audited accuracy and half of it; posterior value of deployable sets at budget 40 and $c = 0$ on the full bank (binomial harm model). Apply attributed uses no prior.

the mean unpatched failure rate. Task effects are shared by the baseline and every patch, so tasks that often fail unpatched do not make patches look harmful, and the common slope β lets patches move every task’s failure rate towards or away from the middle. A patch’s harm is its excess failure probability averaged over the regression tasks, weighted by how often each task was solved among clean training episodes,

$$\rho_k = \sum_t w_t [\sigma(\alpha_t + \beta(\alpha_t - \bar{\alpha}) + \delta_k) - \sigma(\alpha_t)],$$

which respects the support $\rho_k \geq -r_0$ by construction; patches without runs draw δ_k from the population. We sample by Metropolis within Gibbs, four chains, 2,400 retained draws; $\hat{R} \leq 1.006$ for μ , s_d , β and the pooled harm.

Estimates. The unpatched failure rate on solved tasks is 0.146 (retail) and 0.124 (airline). The pooled excess failure rate of the patches is -1.0 points, 90% interval $[-3.4, 1.5]$, on retail and -0.6

$[-3.3, 1.9]$ on airline, and the spread of ρ_k between patches is 2.6 $[0.2, 5.3]$ and 2.1 $[0.2, 4.5]$ points. On retail the slope is negative, $\beta = -0.36$ $[-0.51, -0.17]$: patches tend to raise failures on tasks that rarely fail and lower them on tasks that often fail; on airline it is 0.03 $[-0.21, 0.31]$.

Posterior predictive check. On the patches with 20 runs, the observed standard deviation of failure rates is 0.076 (retail) and 0.066 (airline); data replicated from the fitted model give 0.079 and 0.072, with $P(\text{replicated} \geq \text{observed}) = 0.62$ and 0.79. The number of patches without a failure (4 and 7 observed, 4.2 and 6.2 replicated) and the largest failure rate (0.30 and 0.30 observed, 0.35 and 0.31 replicated) are also reproduced. An earlier analysis used a normal approximation, $y_k = m_k/n_k - r_{0,k} \sim N(\rho_k, v_k)$ with $\rho_k \sim N(\mu, \tau^2)$, which predicted more spread between patches than observed (0.092 and 0.081, $P = 0.95$ and 0.97); its posterior columns in Table 10 and its variants in Tables 3 and 14 are kept as sensitivity analyses.

Sensitivity. Table 14 shows why the harm model matters on retail and hardly on airline. The binomial model values the attributed set at 1.43 oracles on retail; the normal model, which over-predicts the spread between patches and therefore the harm a seven-patch set may carry, values it at 0.77, and its variants range from -1.82 (no pooling) to 1.12 (complete pooling). On airline all models but no pooling give 0.63–0.86. Across models, the best of the five verifying policies beats applying everything in 33–82% of draws on retail and 43–47% on airline.

Budget (episodes)	τ^2 retail (development)						τ^2 airline (held out)					
	significant harm			+ $c_w=0.02$			significant harm			+ $c_w=0.02$		
	10	40	80	10	40	80	10	40	80	10	40	80
Apply attributed	0.84	0.84	0.84	0.16	0.16	0.16	0.60	0.60	0.60	0.15	0.15	0.15
Replay-each	0.78	0.74	0.71	0.14	0.33	0.36	0.43	0.09	0.10	0.07	-0.12	-0.08
HarnessFix-style, per patch	0.52	0.51	0.52	0.52	0.42	0.30	0.18	0.40	0.43	0.18	0.37	0.31
HarnessFix-style, bundle	0.11	0.21	0.26	-0.21	-0.31	-0.31	0.34	0.53	0.52	0.07	0.12	0.11
Uncertainty sampling	0.42	0.50	0.31	0.01	0.23	0.01	0.56	0.50	0.51	0.28	0.25	0.25
Uncertainty + checks	0.14	0.47	0.48	-0.25	0.25	0.24	0.55	0.50	0.53	0.27	0.29	0.36
CARVE, replay only	0.01	0.18	0.14	-0.48	-0.26	-0.39	0.22	0.35	0.48	-0.16	-0.02	0.12
CARVE	0.33	0.32	0.52	-0.01	0.19	0.47	0.55	0.34	0.25	0.30	0.23	0.19
CARVE, measured check	0.01	0.18	0.14	-0.48	-0.26	-0.39	0.25	0.43	0.54	-0.09	0.07	0.20
CARVE, robust check	0.34	-0.03	0.02	-0.15	-0.49	-0.48	0.22	0.34	0.48	-0.12	-0.01	0.14
Net-effect (ours)	0.24	0.52	0.66	0.09	0.41	0.52	0.26	0.39	0.46	0.14	0.31	0.36

Table 9: Sensitivity analysis: as Table 6, but accepted patches are charged only their significant measured regressions (one-sided binomial $p < 0.05$ against r_0), mean over 200 resampling seeds of the original bank. This scoring sets uncertain harm to zero and so favours policies that apply many patches.

Policy (budget 40)		Posterior, $c_w=0.02$		Sensitivity: significant harm only						
		Value [90%]	$P(> \text{apply})$	c_w	c_a	Split	Natural	No step limit	One per comp.	No repl.
Retail	Apply attributed	0.44 [-1.68, 2.40]	—	0.16	0.01	-0.46	-1.25	-0.61	0.47	0.16
	HarnessFix-style, per patch	0.70 [0.04, 1.38]	0.57	0.42	0.34	0.12	-0.82	-0.27	0.42	0.54
	HarnessFix-style, bundle	0.39 [-1.11, 1.89]	0.44	-0.31	-0.42	-0.80	-1.99	-0.83	-0.08	-0.45
	Uncertainty sampling	0.87 [-0.50, 2.15]	0.69	0.23	0.09	-0.13	-1.10	-0.64	0.29	0.28
	CARVE	0.75 [-0.09, 1.65]	0.59	0.19	0.07	0.04	-0.89	-0.25	0.20	0.51
	Net-effect (ours)	0.80 [0.06, 1.52]	0.60	0.41	0.32	0.23	-0.86	-0.35	0.41	0.52
Airline	Apply attributed	0.61 [-0.13, 1.38]	—	0.15	-0.08	-0.41	-0.07	-0.03	0.46	0.15
	HarnessFix-style, per patch	0.44 [0.06, 0.86]	0.26	0.37	0.25	0.17	0.28	0.20	0.38	0.38
	HarnessFix-style, bundle	0.55 [-0.12, 1.25]	0.15	0.12	-0.09	-0.30	-0.09	-0.04	0.39	0.10
	Uncertainty sampling	0.66 [0.06, 1.32]	0.66	0.25	0.01	0.04	0.05	0.05	0.41	0.26
	CARVE	0.36 [-0.09, 0.83]	0.16	0.23	0.08	-0.03	0.24	0.22	0.25	0.23
	Net-effect (ours)	0.65 [0.06, 1.26]	0.58	0.31	0.14	0.17	0.23	0.13	0.35	0.26

Table 10: Uncertainty and robustness at budget 40 (fractions of the repair-only oracle). *Posterior*: deployable sets as in Table 6, mean and 90% interval over the posterior of the earlier normal harm model (Appendix J) and the replay bootstrap, and the probability of beating Apply attributed in the same draw. Sensitivity columns charge significant regressions and $c_w = 0.02$ per wrong change unless stated: c_a , 0.02 per applied change instead; *Split*, policies see a random half of every outcome list and adjudication uses the other half (5 splits); *Natural*, values reweighted to the category shares of natural failures; *No step limit*, the step-limit component removed from the bank; *One per comp.*, the accepted set reduced to one patch per component as deployed; *No repl.*, policies draw bank outcomes without replacement.

K Held-Out Ablations

With a margin of 5 points, chosen after the fact, applying the attributed patches is non-inferior to 9 of the 10 verified sets on retail (all but CARVE with replays only) and to the two HarnessFix-style sets on airline. Single patches show the same problem: leaving the step-limit patch out of the attributed set lowers retail success by 3.3 points (s.e. 4.2) where its replays predict 4.9, and leaving out each policy patch lowers it by 0.0 to 3.3 points where replays predict 0.0 to 0.5, each within about one standard error. The shortfall in retail’s predicted level is therefore spread over the policy patches rather than located in one, and no single patch’s deployed effect is resolved. These runs were made on retail, on

the same held-out episodes as Table 1 (Tables 15 and 16) or on new clean episodes (Table 17). Differences are paired and task-clustered, with 90% intervals. Leaving any one of the seven attributed patches out lowers success by 0.00 to 0.03, and the sum of these contributions is not distinguishable from the whole set’s effect, so the runs neither show nor exclude interactions. On clean episodes, applying everything gains +0.04 [-0.03, 0.10] over the unpatched agent. The bundle gate is ahead of it by +0.04 [0.01, 0.08], but three of its five seed sets are identical to the attributed set and one is empty, so the difference is not an effect of verification; on the 72 slots where it ran the identical set the difference is +0.03 [-0.04, 0.09], which indicates the

Policy (budget 40)		per wrong change c_w				per applied change c_a			
		0	0.01	0.02	0.05	0	0.01	0.02	0.05
Retail	Apply attributed	0.84	0.50	0.16	-0.85	0.84	0.43	0.01	-1.23
	Replay-each	0.74	0.53	0.33	-0.30	0.74	0.48	0.22	-0.57
	HarnessFix-style, per patch	0.51	0.46	0.42	0.28	0.51	0.42	0.34	0.09
	HarnessFix-style, bundle	0.21	-0.05	-0.31	-1.09	0.21	-0.11	-0.42	-1.38
	Uncertainty sampling	-1.90	-2.46	0.23	0.05	-1.90	-2.57	0.09	-0.17
	Uncertainty + checks	-1.57	-1.95	0.25	0.25	-1.57	-2.06	0.12	0.01
	CARVE, replay only	-0.07	-0.49	-0.26	0.00	-0.07	-0.60	-0.44	-0.25
	CARVE	-0.19	-0.19	0.19	0.28	-0.19	-0.27	0.07	0.14
	CARVE, measured check	-0.07	-0.49	-0.26	0.00	-0.07	-0.60	-0.44	-0.25
	CARVE, robust check	-0.05	-0.70	-0.49	0.01	-0.05	-0.81	-0.66	-0.26
	Net-effect (ours)	-1.02	-0.58	0.41	0.63	-1.02	-0.67	0.32	0.44
Airline	Apply attributed	0.60	0.37	0.15	-0.53	0.60	0.26	-0.08	-1.09
	Replay-each	0.09	-0.02	-0.12	-0.43	0.09	-0.08	-0.25	-0.76
	HarnessFix-style, per patch	0.40	0.39	0.37	0.32	0.40	0.33	0.25	0.02
	HarnessFix-style, bundle	0.53	0.32	0.12	-0.50	0.53	0.22	-0.09	-1.02
	Uncertainty sampling	0.49	0.09	0.25	0.19	0.49	-0.04	0.01	-0.25
	Uncertainty + checks	0.44	0.01	0.29	0.21	0.44	-0.12	0.08	-0.13
	CARVE, replay only	0.48	0.08	-0.02	0.06	0.48	-0.07	-0.26	-0.44
	CARVE	0.71	0.33	0.23	0.10	0.71	0.22	0.08	-0.07
	CARVE, measured check	0.56	0.10	0.07	0.11	0.56	-0.04	-0.17	-0.38
	CARVE, robust check	0.63	0.21	-0.01	0.08	0.63	0.06	-0.26	-0.42
	Net-effect (ours)	0.60	0.10	0.31	0.29	0.60	-0.04	0.14	0.05

Table 11: Value at budget 40 as a fraction of the repair-only oracle, with significant regressions charged, as the change cost varies. Policies that use a cost internally are told the same c (the net-effect method always charges it per applied change, the others per wrong change). At $c = 0$ the uncertainty-based policies accept almost every cell with a positive posterior and pick up patches with significant regressions.

Regression term	Retail			Airline		
	10	40	80	10	40	80
As used (clip, rescale)	0.70	0.90	1.08	0.63	0.70	0.77
No clip	0.65	0.38	0.52	0.78	0.82	0.81
No rescale	0.70	0.90	1.08	0.63	0.70	0.77
Neither (Eq. 2)	0.63	0.40	0.56	0.79	0.82	0.81

Table 12: Net-effect method with its regression term as used (clipped at zero, scaled by $1/(1 - r_0)$) and without either, posterior value of deployable sets at $c = 0$ over 40 seeds.

run-to-run noise of these comparisons. Unpatched clean success was 0.82 both before and after the new runs.

L Development Variants

On the retail bank we tried, before any airline evaluation of them:

- CARVE with two null replays per category before trusting its replays (per-category b);
- CARVE with a regression gate of two runs per accepted patch;
- both combined;
- an evidence rule that requires a positive likelihood ratio from data;

- tempered and flat priors;

- a Bayesian regression gate.

With control replays and the gate combined, CARVE gained 0.68 and 0.60 of the oracle at budgets 10 and 20, above plain CARVE (0.42, 0.48), but fell below it at 40 and 80 (0.47 vs. 0.57 and 0.56). It never beat the per-patch HarnessFix-style rule (0.72, 0.73). Post hoc on airline it was worse than plain CARVE (0.19 vs. 0.31 at budget 10). The other variants were worse everywhere; the evidence rule and the Bayesian gate accepted almost nothing (gain ≤ 0.12). These numbers use the fixed change cost c without regressions and the bank before the regression runs were extended. The net-effect method of Appendix D was developed afterwards, on retail, from the diagnosis that CARVE lacked per-category spurious recovery and evidence for regressions.

M ShopDesk Details

ShopDesk has six deterministic tools (orders, products, policies, refunds), answers checked exactly (refund amounts, counts, totals), and five injected faults: a tool description misstating price units, a list tool whose description omits pagination, a

Judge	Cost per check		vs. single replay	
	Tokens	Pro $2\times / 4\times$	Break-even	Ratio [95% CI]
<i>Retail</i>				
Bank judge (analyst, full trajectory)	0.29	0.43 / 0.70	0.14	0.47 [0.25, 0.70]
Analyst, failing step only	0.30	0.45 / 0.76	0.10	0.32 [0.13, 0.51]
Agent model, full trajectory [†]	0.67	0.67 / 0.67	0.11	0.16 [0.01, 0.31]
Agent model, failing step only	0.28	0.28 / 0.28	0.10	0.35 [0.13, 0.57]
<i>Airline</i>				
Bank judge (analyst, full trajectory)	0.22	0.33 / 0.54	0.14	0.62 [0.33, 0.91]
Analyst, failing step only	0.23	0.34 / 0.57	0.10	0.43 [0.17, 0.69]
Agent model, full trajectory [†]	0.58	0.58 / 0.58	0.24	0.40 [0.10, 0.70]
Agent model, failing step only	0.22	0.22 / 0.22	0.09	0.39 [0.16, 0.62]

Table 13: Single-step checks with four judges. Cost is per check as a fraction of a full replay in uncached input plus output tokens, including regenerating the failing step (0.15 of a replay on retail, 0.12 on airline); “Pro $2\times / 4\times$ ” prices the analyst model’s tokens at twice or four times the agent model’s, which affects only analyst-model judges (replays use the agent model). Break-even is the cost ratio below which checks decide an edge at lower expected cost than replays (Proposition 4, with the domain’s mean spurious recovery and median repair rate of effective edges, averaged over true and false edges); Ratio is break-even divided by cost, so values above 1 favour checks. *vs. single replay*: sensitivity and false-positive rate conditional on the paired replay’s outcome (360 stratified pairs; Table 4 gives the bank judge’s on all pairs). *vs. patch replay rate*: $z = \epsilon + (s - \epsilon)p_k$ fitted by moments with the binomial noise of each patch’s 20-replay rate removed, which is the observation model of Proposition 4; the bank judge row uses all 2,160 pairs, the others their 360 pairs. Intervals bootstrap source trajectories (single replay) or patches (patch rate) together with null replays. [†]40 random pairs per domain.

Harm model	Retail					Airline				
	μ	Apply	Best	P	VPI	μ	Apply	Best	P	VPI
Task-level binomial (primary)	−0.01	1.43	1.39	0.33	1.18	−0.01	0.66	0.65	0.45	0.86
Hierarchical normal	−0.00	0.77	1.05	0.63	2.02	−0.01	0.73	0.71	0.46	0.69
Complete pooling ($\tau = 0$)	−0.00	1.12	1.02	0.36	0.87	−0.01	0.73	0.71	0.46	0.69
Spread doubled (2τ)	−0.00	0.27	0.96	0.70	2.51	−0.01	0.73	0.71	0.46	0.69
No pooling	0.04	−1.82	0.20	0.82	4.65	0.04	0.39	0.37	0.43	1.15
Tasks resampled	−0.00	−0.52	0.58	0.65	3.31	−0.00	0.63	0.62	0.45	0.87
No support cap	−0.00	0.82	1.09	0.64	9.48	−0.01	0.74	0.71	0.45	1.06
Not deployable	−0.00	0.64	1.14	0.61	2.15	−0.01	0.86	0.86	0.47	0.56

Table 14: Sensitivity of the posterior to the harm model, $c = 0$, budget 40: mean harm μ (for the binomial model, the pooled excess failure rate), value of Apply attributed and of the best of five verifying policies (both HarnessFix-style gates, uncertainty sampling, CARVE, net-effect), the probability that it beats Apply attributed, and the total value of perfect information over Apply attributed, as fractions of the oracle. Rows below the first use the earlier normal approximation; airline has $\tau = 0$ there, so pooling and doubling τ change nothing.

Set (seed 0)	With – without step patch			Without – unpatched	
	All	No step fault	Step fault	No step fault	
Apply attributed	0.03 [−0.04, 0.10]	−0.03 [−0.10, 0.04]	0.58 [0.37, 0.80]	0.09 [0.01, 0.18]	
HarnessFix-style, bundle	0.03 [−0.04, 0.10]	−0.03 [−0.10, 0.04]	0.58 [0.37, 0.80]	0.09 [0.01, 0.18]	
Uncertainty sampling	0.01 [−0.07, 0.08]	−0.02 [−0.09, 0.06]	0.25 [−0.07, 0.57]	0.06 [−0.04, 0.15]	
CARVE	0.03 [−0.04, 0.11]	0.00 [−0.07, 0.07]	0.33 [0.07, 0.59]	0.05 [−0.03, 0.13]	
Net-effect (ours)	0.05 [−0.02, 0.12]	−0.01 [−0.08, 0.06]	0.58 [0.34, 0.83]	0.02 [−0.07, 0.10]	

Table 15: Removing the step-limit patch from each policy’s seed-0 set (the bundle gate’s seed-0 set equals the attributed set). 108 episodes without and 12 with the step-limit fault.

Patch left out of Apply attributed	All	No step fault
CFG.max_steps	0.03 [−0.04, 0.10]	−0.03 [−0.10, 0.04]
POL.cancel	0.03 [−0.04, 0.10]	0.04 [−0.04, 0.12]
POL.exchange	0.03 [−0.04, 0.09]	0.03 [−0.05, 0.11]
POL.generic	0.03 [−0.04, 0.09]	0.01 [−0.05, 0.07]
POL.intro	0.02 [−0.05, 0.08]	0.02 [−0.05, 0.08]
POL.modify	0.03 [−0.05, 0.10]	0.04 [−0.05, 0.12]
POL.return	0.00 [−0.06, 0.06]	−0.01 [−0.08, 0.06]
Whole set vs unpatched	0.12 [0.05, 0.19]	0.06 [−0.01, 0.14]
Whole set – sum of parts	−0.04 [−0.35, 0.27]	−0.03 [−0.38, 0.33]

Table 16: Leave-one-out on the attributed set: success of the whole set minus the set without the named patch.

format instruction that rounds to whole dollars, a planning instruction that skips the policy lookup, and a tight step limit. Over three seeds, the analyst’s attribution accuracy on injected failures was 0.75, 0.64 and 0.77. ShopDesk was evaluated on-

line, without a bank: each policy ran against the live agent with fault-based ground truth (an edge is real when at least 30% of a category’s failures come from that component’s fault). Table 18 reports edge

	Old	New	Pooled	– Unpatched	– Apply
Unpatched	0.82	0.82	0.82	–	–0.04 [–0.10, 0.03]
Apply attributed	0.87	0.85	0.85	0.04 [–0.03, 0.10]	–
Net-effect (ours)	0.83	0.87	0.86	0.04 [–0.01, 0.09]	0.00 [–0.05, 0.06]
HarnessFix-style, bundle	0.87	0.91	0.90	0.08 [0.03, 0.14]	0.04 [0.01, 0.08]
HarnessFix-style, per patch	0.90	0.85	0.86	0.05 [–0.01, 0.10]	0.01 [–0.06, 0.08]

Table 17: Clean held-out episodes: 38 old (Table 1) and 120 new (40 tasks $\times$ 3), where new episode e uses the policy’s seed- $(e \bmod 5)$ set. Differences pool both, paired by episode or slot and clustered by task.

F1 and held-out success. Large harms do occur: a formatting patch that passed its target-category replays made 26% of held-out tasks answer N/A (success 0.89 $\rightarrow$ 0.73).

N Study 2: RRSI Port and Registered Analysis

Code. RRSI’s loop is vendored from its public repository under the Apache 2.0 licence, with the upstream copyright headers kept. We changed the model backend (calls go to DeepSeek’s Anthropic-compatible endpoint, with thinking disabled and token usage recorded per role), and one rule: a proposal that ends with no file change and no declared edit is bounced back to the proposer like an aborted one, at most three times, instead of ending the round with no candidate. RRSI’s selection rule is used unchanged: a candidate is admissible if its evolve success is at most δ below the best so far and either exceeds the incumbent’s by more than δ at an increase in running cost within $\beta_0 + \beta_1 \Delta S$, or lies within the band with a positive weighted sum of gain, cost saving and novelty (with RRSI’s weights, gain counts 0 there, so only a cheaper or untried kind of change passes); a harness with errors in more than 2% of trials is rejected; the admissible candidate with the highest success wins. RRSI’s own domains (coding, workspace, engineering design) are not used. Our τ^2 adapter implements RRSI’s domain interface: a frozen episode driver that uses τ^2 ’s databases, tools, policies, tasks and user-simulation guidelines, a starting harness with four hooks (choose the next action, check a tool call before it runs, render a tool result, rebuild per-episode state from a recorded prefix), and optional mechanisms the proposer may use: bounded extra model calls, harness-side tools and skill files. Memory across episodes is not allowed.

Harness check. At temperature 0 our episode loop matches τ^2 ’s own orchestrator on 60 test tasks $\times$ 3: 90.0% against 90.8% success on retail (paired difference -0.8 points, $[-5.8, 3.3]$) and 71.7%

against 71.7% on airline. Scoring τ^2 ’s natural-language assertions as well changed the reward of 2 of about 1,000 episodes, so we keep Study 1’s reward (database state and required information).

Leakage screening. Besides RRSI’s LLM critic, a pattern check rejects any added line that contains a task’s user id, order or reservation id, item or flight number or customer e-mail, refers to the grader (gold actions, database hashes, evaluation criteria) or to τ^2 ’s task data, imports the benchmark or our experiment code, or writes files.

Evidence collection per round. *References:* for every evolve task the incumbent failed in RRSI’s full evaluation, its worst trial, at most 20 tasks (a seeded random subset beyond that). The start step is the earliest step the round’s failure digest cites; without a digest, the first state-changing call that is not a gold action; otherwise the first step. Each reference is replayed 4 times with each measured candidate and 4 times with the incumbent (null replays). *Judge:* DeepSeek-V4-Pro reads the candidate’s diff and declared edits, up to 14 of the incumbent’s failed trials of the round with their grading block, and one line per solved task, and returns a probability of repair for each failed task and an expected number of broken trials; its estimate is (expected repairs – expected breaks) / evolve trials. *Deployment:* every incumbent and measured candidate on the held-out tasks, retail 6 and airline 12 trials per task (4 and 8 as registered; see below), the base and final harnesses on 10 (retail) and 12 (airline) trials per task.

Estimators. With f the incumbent’s evolve failure share, $sample@b$ is the mean paired difference on b random (task, trial) pairs of the full evaluations; $replay@b$ is $f \cdot \widehat{fix}_c$ from b candidate replays spread evenly over the references; $replay-null@b$ is $f(\widehat{fix}_c - \widehat{fix}_{null})$ from $b/2$ replays of each kind; $net@b$ uses $b/4$ candidate and $b/4$ null replays and $b/2$ regression pairs, on tasks the incumbent solved, and adds $(1 - f)$ times their mean paired difference; $net@b$ with *prior* shrinks the candidate’s repair rate to a Beta prior with mean 0.41 or 0.29, 0.7 or 0.2 and two pseudo-trials; *full* is RRSI’s own decision. Every estimate goes through RRSI’s selection rule with the evidence’s own noise band, twice its standard error, and the evidence’s measured token cost; the judge has no noise band. Each rule’s per-round score is the deployment gain of its choice, averaged over 400 random subsamples of the evidence.

Method	Edge F1 at budget				Held-out success @60	Single checks
	10	20	30	60		
LLM attribution only	0.61 $\pm$ 0.04	0.61 $\pm$ 0.04	0.61 $\pm$ 0.04	0.61 $\pm$ 0.04	0.98 $\pm$ 0.00	0
Replay-each (do-then-verify)	0.62 $\pm$ 0.03	0.63 $\pm$ 0.03	0.67 $\pm$ 0.05	0.75 $\pm$ 0.03	0.99 $\pm$ 0.00	0
Uncertainty sampling	0.55 $\pm$ 0.04	0.57 $\pm$ 0.04	0.58 $\pm$ 0.05	0.66 $\pm$ 0.04	0.99 $\pm$ 0.00	0
Uncertainty + cheap checks	0.52 $\pm$ 0.07	0.56 $\pm$ 0.07	0.53 $\pm$ 0.09	0.53 $\pm$ 0.08	0.96 $\pm$ 0.03	57
CARVE, full replay only	0.61 $\pm$ 0.07	0.61 $\pm$ 0.06	0.59 $\pm$ 0.04	0.64 $\pm$ 0.03	0.91 $\pm$ 0.09	0
CARVE (default check model)	0.59 $\pm$ 0.06	0.57 $\pm$ 0.05	0.61 $\pm$ 0.05	0.68 $\pm$ 0.03	0.98 $\pm$ 0.01	4
CARVE, calibrated check	0.59 $\pm$ 0.06	0.58 $\pm$ 0.05	0.60 $\pm$ 0.05	0.66 $\pm$ 0.04	0.89 $\pm$ 0.08	0
CARVE, robust check model	0.58 $\pm$ 0.07	0.61 $\pm$ 0.06	0.61 $\pm$ 0.04	0.69 $\pm$ 0.01	0.99 $\pm$ 0.01	0
Unpatched agent	0.59 $\pm$ 0.04					

Table 18: ShopDesk (3 seeds, mean $\pm$ s.e.). Held-out success after applying the patches accepted at budget 60 to 120 held-out episodes.

Evidence	Episodes	Estimate of ΔS
None	0	accept a candidate (at random if two)
LLM judge	0	predicted fixes — predicted breaks
Sample@ b	b	paired difference on b evolve episodes
Replay@ b	b	failure share $\times$ repair rate
Replay—null@ b	b	failure share $\times$ (repair — null rate)
Net@ b	b	replay—null — success share $\times$ regressions
Net@ b , prior	b	net@ b with a Beta prior on the repair rate
Full (RRSI)	148 / 60	paired difference on all evolve episodes

Table 19: Evidence compared in Study 2, per candidate, with $b \in \{10, 40, 80\}$; full evaluation costs 148 (retail) or 60 (airline) episodes. The priors have means 0.41 or 0.29 (audited analyst accuracy), 0.7 and 0.2. Every rule decides with RRSI’s selection rule.

The registered analysis applies RRSI’s error guard, which rejects a harness that raises errors in more than 2% of trials, only through RRSI’s own decision, not to the other rules’ evidence; §5 reports the alternative post hoc.

Registration of r1 and changes after it. The pre-registration (verify/PREREGISTRATION.md), the analysis code and the run scripts were committed before the first held-out episode; domains, splits, models, temperatures, T , m , k , budgets, evidence definitions, the decision rule, the margin and the endpoints did not change. The registration file logs four changes. (i) Before any held-out episode, a bug fix: the failure share f was computed from the replay references, which are capped at 20 tasks, instead of from all failed trials of the incumbent’s evaluation. (ii) Also before any held-out episode, Net(N), a registered secondary outcome, was implemented as the mean over rounds of $N(v\Delta p - \Delta c_{\text{run}}) - C_{\text{verify}}$ against keeping the incumbent, with $\Delta C_{\text{change}} = 0$ because every rule sees the same candidates. (iii) After six rounds, net@40 and full evaluation chose differently in 58% of them, above the 40% of the power analysis,

and held-out trials per harness were raised from 160 to 240 (retail 4 to 6, airline 8 to 12 per task, airline endpoints 10 to 12); harnesses already deployed were topped up. The registration assumed this rate would be known before any held-out episode ran; in fact the first rounds had been deployed, and a progress log had shown a few held-out success rates while the run was monitored, although no held-out analysis was run. (iv) Over all 40 rounds the rate was 26.5%, for which the registered 160 episodes would have sufficed; we report 240 as primary and 160 as a sensitivity analysis, with the same conclusions. The analyses without the broken candidate, with the error check applied to every rule, with the execution-check baselines, with blocks of rounds that share an incumbent and of detectable differences, and the rules that verify one candidate or verify sequentially, are post hoc.

Registration of r2 and r3. After r1’s results, we registered r2 and r3 (verify/PREREGISTRATION_E1R.md) before any of their episodes ran, with the analysis of §5. The analysis code (verify/e1r.py) was committed while they ran and tested on r1 only. The sequential rules and their tests were added as an addendum before any decision value of r2 or r3 was computed. The operations log records the following. In r2’s third retail round, RRSI’s digester received a string where it expected an object and the driver stopped; the fix returns such a reply to the digester as an error and changes nothing else. The container restarted twice, and every run resumed from its last settled step, reusing the reports, drafts and evaluations it had written, so no completed step was sampled again. While the second restart was checked, progress logs displayed the held-out success rates of a few

Evidence	b	Registered analysis		Common error check		Accepts	Cost per round episodes / equiv.	Net(10^4) $v = 10$
		D	[90%]	D	[90%]			
None	0	-1.87	[-4.61, +0.25]	-1.87	[-4.61, +0.25]	0.90	0 / 0	-1,904
LLM judge	0	-1.72	[-4.42, +0.28]	-1.72	[-4.42, +0.28]	0.72	0 / 0	-1,763
None + check	10	-	-	-0.58	[-1.94, +0.79]	0.88	12 / 12	-652
Judge + check	10	-	-	-0.43	[-1.61, +0.72]	0.70	12 / 12	-512
Full (RRSI)	all	+0.22	[-0.06, +0.62]	+0.22	[-0.06, +0.62]	0.12	123 / 123	78
Sequential full	$\leq$ all	-	-	+0.15	[-0.05, +0.42]	0.10	65 / 65	75
Sequential sample	$\leq$ 80	-	-	+0.12	[-0.19, +0.51]	0.17	43 / 43	77
Sample	10	-0.34	[-0.99, +0.21]	-0.04	[-0.44, +0.35]	0.30	12 / 12	-69
Replay	10	-0.43	[-1.27, +0.37]	-0.37	[-1.20, +0.45]	0.54	12 / 9	-423
Replay-null	10	-0.05	[-0.26, +0.13]	-0.05	[-0.26, +0.13]	0.08	10 / 8	-69
Net	10	-0.50	[-1.43, +0.23]	-0.07	[-0.57, +0.41]	0.37	11 / 10	-101
Sample	40	+0.04	[-0.38, +0.45]	+0.04	[-0.38, +0.45]	0.28	47 / 47	-6
Replay	40	-1.67	[-4.35, +0.26]	-0.40	[-1.60, +0.78]	0.80	42 / 33	-499
Replay-null	40	-0.07	[-0.35, +0.15]	-0.07	[-0.35, +0.15]	0.10	42 / 34	-125
Net	40	-0.07	[-0.52, +0.36]	-0.04	[-0.50, +0.38]	0.30	44 / 40	-101
Sample	80	+0.08	[-0.35, +0.54]	+0.08	[-0.35, +0.54]	0.26	82 / 82	3
Replay	80	-1.69	[-4.41, +0.27]	-0.40	[-1.59, +0.79]	0.80	47 / 37	-496
Replay-null	80	-0.12	[-0.59, +0.27]	-0.12	[-0.59, +0.27]	0.14	74 / 59	-210
Net	80	-0.02	[-0.49, +0.46]	-0.02	[-0.49, +0.46]	0.27	88 / 81	-119

Table 20: Study 2, first trajectory (r1): decision value D of each rule, the held-out gain of its choice over the incumbent in points per round (keeping the incumbent scores 0), over 40 rounds of both domains, with a 90% bootstrap interval over rounds and tasks (Table 2 gives intervals from blocks of rounds). Registered: RRSI’s harness-error check (reject above 2% errors) only inside full evaluation. Common error check (post hoc): the check applied to every rule’s own episodes; + *check* rules run 10 episodes of each candidate only for the check; sequential rules stop a candidate’s episodes early (post hoc, Appendix N). Accepts, cost and Net(10^4) (Eq. 1 against keeping the incumbent, $N = 10^4$ deployed episodes, a success worth $v = 10$ episodes, in episodes) are for the common check. No candidate was measured in 4 of the 40 rounds. Full: RRSI’s own decision. Best value in bold.

rounds’ harnesses (r2 retail round 10, r2 airline 12, r3 retail 9, r3 airline 15). Nothing was decided from them, and the analysis was fixed before.

Thinking proposer (r4). A fourth trajectory per domain, registered as descriptive, runs the proposer, analyst, digester and critic with thinking enabled (8,000 tokens). With thinking, a draft took 1–1.6 hours and 4 of the first 5 drafts after round 0 ended at the proposer’s turn limit without a proposal. Retail’s noise band was calibrated at 17.7 points from base evaluations of 89.2%, 78.4% and 89.2% (r1–r3: 1.1–4.0), so its acceptance rule was not comparable; it was stopped after two rounds. Airline ran ten rounds, after which the base and final harnesses were deployed; we compare it with the first ten airline rounds of r1–r3. Of its 20 drafts, 14 ended without a proposal (r1–r3: 2–6), so it measured 6 candidates (13–15) and accepted 2 (3–4). All six measured candidates had a positive deployment gain, on average +3.5 points (r1–r3: +0.2 to +1.0, with 40–54% positive), although with a single-candidate MDE of 8.5 points none is resolved on

its own. Because nearly every candidate helped, the rules hardly differed: decision values per round lay between +0.85 (sequential full) and +1.17 (no verification), with full evaluation at +0.92 (r1–r3: +0.29 to +0.87) and every 90% interval reaching from about 0 to +6 or more. After ten rounds the incumbent gained +9.2 [+3.8, +15.4] points held out over the base harness (r1–r3 after ten rounds: +8.8, +8.8, +2.9) and +20.0 on the evolve set. Thinking raised the loop’s model cost per round by a factor of 3.4–3.9 over r2 and r3. On this one trajectory a stronger proposer drafted fewer but better candidates, which left little for verification to decide; one trajectory cannot separate that from the variation between trajectories.

Sequential rules. Designed post hoc on r1 and registered for r2 and r3, a sequential rule plans N paired episodes per candidate (RRSI’s full evaluation, or 80) and runs them in batches of 10 in random order. After n episodes with mean paired difference m it drops the candidate if its harness errors exceed 2% of N , or if the predic-

tive probability that RRSI’s rule admits it after N episodes is below $\gamma = 0.05$. For that probability the final ΔS is normal with mean m and variance $\left(\frac{N-n}{N}\right)^2 s^2 \left(\frac{1}{n} + \frac{1}{N-n}\right)$, where s^2 is the larger of the running variance of the paired differences and $S_t(1 - S_t)$; the cost change is held at its running estimate and the novelty count is known. Candidates that are not dropped complete the N episodes and are decided as without stopping; without the predictive stop, sequential full evaluation makes every decision full evaluation makes. With $\gamma = 0.01$ and 0.10 , sequential full evaluation gains $+0.17$ at 81 and $+0.13$ at 55 episodes per round on $r1$, and $+0.03$ at 100 and $+0.02$ at 68 on $r2$ and $r3$ ($+0.02$ at 81 with $\gamma = 0.05$). Fully evaluating only the candidate the judge prefers gains $+0.00$ on $r2$ and $r3$ at 97 episodes, and a random one $+0.03$.

Classifying candidates. We call a candidate a code change if it edits the harness’s agent, tool or check modules, and a text change if it edits only the system prompt or skill files. Code changes rarely lengthened replays: 88 of $1,222$ replays of code changes went live before the requested step, all for 4 of 31 candidates, and none of 640 replays of text changes did. RRSI’s own component labels rely on file-name patterns that do not match our harness, and called most code changes prompt edits. The labels also feed RRSI’s record of which kinds of change have been tried, which it uses to point the proposer at untried kinds and in the novelty term of its selection rule, so the loop explored as RRSI would on a harness whose files its patterns do not recognise; one candidate declared as a harness-side tool was labelled a configuration change. For $r2$ and $r3$ the domains’ signals see git’s python diff driver and label code in the agent module or in a new harness module as control flow; on $r1$ ’s 47 candidates this gives 24 control-flow and 18 prompt labels instead of 2 and 38 , and no code change is labelled prompt only. With these labels the 107 candidates of $r2$ and $r3$ were labelled control flow 44 times, prompt 32 , rendering of tool results 26 , skill 10 , harness-side tool 4 , extra model call 4 , context 3 and configuration 2 (a candidate can carry several labels). As on $r1$, code changes rarely lengthened replays (316 of $3,236$, for 12 of 75 candidates), and replays of text changes never did.

Planned and actual cost. The registered estimate was, per domain, the RRSI trajectory (three base evaluations and up to 40 candidate evalu-

ations), about $2,000$ – $2,700$ targeted replays and about 48 harnesses deployed on 160 held-out episodes each: about $30,000$ episodes in all, plus the search and judge models, about $\$70$. The study used $21,151$ episodes (retail $12,428$, airline $8,723$): $5,556$ in RRSI’s evaluations, $3,515$ replays and $12,080$ held-out episodes, and about $\$37$ at third-party list prices, $\$10$ of it for DeepSeek-V4-Pro. Only 47 of 80 drafts were measured, so fewer replays and deployments were needed than planned. $r2$ and $r3$ were quoted at about $42,000$ episodes and $\$75$, with a stop at $\$110$. They used $46,584$ episodes ($11,480$ in RRSI’s evaluations, $7,824$ replays and $27,280$ held-out episodes) and about $\$83$, $\$20$ of it for DeepSeek-V4-Pro.

O Study 2: Further Results on the First Trajectory

Registered endpoints. For $r1$ the registered primary endpoints are (P1) transfer, the held-out gain of the final harness over the base, and (P2) D at $b = 40$, with three one-sided contrasts, Holm-adjusted at $\alpha = 0.05$: net@40 against full evaluation (non-inferiority, margin 0.75 points per round), net@40 against no verification, and full evaluation against no verification. Its bootstrap resamples rounds and held-out tasks within domain. The registered 160 held-out episodes per harness were raised to 240 after six rounds, as the registration allowed (Appendix N); 160 is a sensitivity analysis.

P2, as registered. Full evaluation had the highest decision value, $+0.22$ points per round $[-0.06, +0.62]$ (Table 20, Appendix N); sample@40 and sample@80 also had small positive point estimates. Net@40, at a third of the cost, had -0.07 ; the non-inferiority difference was -0.28 $[-0.77, +0.17]$, one-sided $p = 0.058$ (Holm 0.17), so neither non-inferiority nor inferiority was shown. Neither rule was shown better than no verification (full $+2.09$, net@40 $+1.80$; Holm $p = 0.17$). With the registered 160 held-out episodes the non-inferiority difference was -0.48 $[-1.03, +0.01]$, $p = 0.185$. The priors moved net@40 by at most 0.01 .

The broken candidate and a common error check. Most of the losses of no verification, the judge and replay without null replays come from one candidate (retail, round 6): a check before item-changing calls that reads an attribute the harness never sets, so the harness raised an error in

half of its evolve episodes and held-out success fell by 51.7 points. RRSI’s critic accepted it and the judge predicted a gain of 1.5 points; 6 of its 48 replays still succeeded, which $\text{replay}@b$ counts as repairs, while 15 of 48 null replays succeeded. The registered analysis gave RRSI’s error check to full evaluation only. To compare evidence rather than error handling, we apply the check to every rule’s own episodes and add two baselines that run ten episodes of each candidate only for the check, then accept a passing candidate at random or by the judge (post hoc; Tables 20 and 2). Every rule that runs episodes then rejects the broken candidate. Full evaluation still has the highest value; its differences from $\text{sample}@40$, $\text{net}@40$ and $\text{replay}-\text{null}@40$ are $+0.18 [-0.22, +0.58]$, $+0.26 [-0.20, +0.75]$ (non-inferiority $p = 0.049$, Holm 0.15) and $+0.29 [-0.06, +0.72]$, and these three differ from each other by at most 0.11. Accepting what passes the execution check loses 0.58 points per round $[-1.94, +0.79]$ (0.43 with the judge), 0.80 $[-0.67, +2.19]$ less than full evaluation: beyond crashes, most candidates still hurt.

Dependence and resolution. Retail’s 20 rounds share 3 incumbents and airline’s 4. Resampling blocks of rounds that share an incumbent widens the standard errors of these contrasts by a factor of 1.3–1.9 (full minus $\text{net}@40$: $[-0.27, +1.00]$), and leaving out any one incumbent moves them by at most 0.20 points. With 80% power at a one-sided 5% level, the design detects per-round differences of 0.60–0.74 points between full evaluation and the $@40$ rules (0.78–1.11 by blocks), 0.29–0.52 among the $@40$ rules and about 2 against the execution-only baseline; a single candidate’s gain is detected only above about 5.3 points (median standard error 2.1). Apart from the broken candidate, gains were too small to replicate: the split-half correlation of candidates’ held-out gains was 0.76 with it and -0.30 without it. Correlations with deployment gain were 0.85 for RRSI’s evolve difference (0.33 without that candidate), 0.03 (0.08) for the judge and 0.25 (0.26) for replays.

Spending fewer episodes (post hoc, exploratory on r1). Fully evaluating only the candidate the judge prefers gains $+0.14$ points per round at 94 episodes, as does a random one. Sequential rules run a candidate’s evaluation in batches of ten in random order and drop it once its errors exceed the check or the predictive probability that RRSI’s rule will admit it falls below 0.05 (Ap-

pendix N). Sequential full evaluation gains $+0.15 [-0.05, +0.42]$ at 65 episodes per round, $0.07 [-0.03, +0.24]$ below full evaluation at about half its cost. Sequential sampling capped at 80 episodes gains $+0.12 [-0.19, +0.51]$ at 43, against $+0.04$ for fixed $\text{sample}@40$ at 47.

Cost. On r1, full evaluation cost 123 episodes per round and the $@40$ rules 33–47 episode equivalents, because a replay used about 0.8 of an episode’s tokens. Weighting tokens by list prices changes no ranking; it adds the judge’s calls, on the larger model, at about 7 episodes per round. Against keeping the incumbent, full evaluation’s $\text{Net}(10^4)$ at $v = 10$ is $+78$ episodes $[-183, +466]$, positive in 58% of resamples; its break-even horizon of about 6,000 episodes is a point estimate, and at $v = 1$ no rule repays its evidence within 10^4 episodes. r1 used about 21,200 episodes and \$37 (Appendix N).

P Study 2 on AppWorld (E3)

Design. Registered before any episode of its loop, E3 runs the loop, evidence and analysis of r2 and r3 on AppWorld 0.1.3 (Trivedi et al., 2024). The agent is DeepSeek-V4-Flash at temperature 0 with AppWorld’s ReAct prompt; each step is one assistant turn whose first Python block runs in a persistent interpreter that calls the apps’ APIs. Evolve tasks are the first 20 training scenarios (60 tasks, two trials each) and held-out tasks are `test_normal` (168 tasks, scenarios disjoint from training); every candidate and incumbent is deployed on two trials per held-out task and the base and final harnesses on four. A task succeeds when all of its unit tests pass. The judge’s prompt calls the agent a code agent; nothing else differs. Base evaluations gave 88.3%, 90.0% and 89.2%, a noise band of 2.4 points. The loop ran $T = 20$ rounds, drafted 40 candidates and measured 29 (16 changed code), and accepted three, two within the band. No candidate’s harness errors exceeded the check. The study used 15,472 episodes and about \$48.

Results. Table 21. Accepting every candidate gained -0.02 points per round and the judge’s choice $+0.18$; candidates’ gains exceeded the single-candidate detectable effect of 3.2 points for 1 of 29, and their split-half correlation was 0.36. Under r1’s registered analysis, with its bootstrap over rounds, $\text{net}@40$ was non-inferior to full evaluation (Holm $p = 0.036$). $\text{Net}(10^4)$ at $v = 10$ was negative for every rule (-160 episodes for full

Evidence	D [90%]	Ep.	— full [90%]	p (Holm)
Full (RRSI)	+0.03 [−0.30, +0.71]	174	—	—
Sequential full	+0.08 [−0.19, +0.72]	108	+0.05 [−0.00, +0.14]	<0.001 (<0.001)
Sequential sample@80	−0.01 [−0.22, +0.31]	58	−0.04 [−0.47, +0.14]	0.019 (0.019)
Sample@40	−0.04 [−0.38, +0.41]	58	−0.07 [−0.49, +0.15]	0.018 (0.053)
Net@40	−0.08 [−0.46, +0.37]	53	−0.11 [−0.59, +0.18]	0.025 (0.053)
Replay—null@40	+0.22 [−0.20, +0.79]	48	+0.19 [−0.47, +0.67]	0.023 (0.053)
Sample@80	−0.05 [−0.38, +0.44]	116	—	—
None + check	−0.02 [−1.10, +1.35]	14	—	—
Judge + check	+0.18 [−0.86, +1.51]	14	—	—

Table 21: E3 (AppWorld, 20 rounds): decision value D under the common error check in points per round, episodes per round, and the registered non-inferiority contrasts with full evaluation (margin 0.75, one-sided p , Holm within each family), with 90% intervals from the block bootstrap.

evaluation, −39 for sequential full evaluation).

Q Margins, Decision Rules, Calibration and Loop Cost

The analyses of this appendix are post hoc and use no new episodes: they recompute the registered decision values from the same evidence and deployments (`verify/posthoc_r7.py`).

Smaller margins and agreement. Table 22 repeats the non-inferiority tests against full evaluation at margins of 0.1, 0.25 and 0.5 points per round, adds keeping the incumbent as a rule with zero cost and zero value, and reports how often each rule makes full evaluation’s choice (for keeping the incumbent, how often full evaluation keeps it). At the registered margin every rule that runs solved tasks passes, and so does keeping the incumbent; at 0.25 points only sequential full evaluation passes on r2+r3 (sequential sampling: $p = 0.056$), and on r1, the design set, it passes from 0.5. On r4, where full evaluation gained, keeping the incumbent fails at every margin and the sequential rules pass at 0.25.

Evidence or decision rule. In the registered analysis full evaluation is RRSI’s own decision, with the noise band calibrated from three base evaluations, while every other rule uses twice its estimate’s standard error, and RRSI admits a candidate within its band if it is cheaper or of an untried kind. Table 23 decides full evaluation (and sequential full evaluation) like the other rules, with its own band, and then decides every rule by one criterion: admit only if the estimate exceeds twice its standard error and the cost condition holds, with no within-band branch. The judge and no verification have no standard error and are un-

changed. With their own band, full evaluation’s value hardly changes. With one criterion every rule that runs episodes accepts a candidate in at most 3% of rounds (replay—null@40: 7%), and its value moves towards zero; replay—null@40 remains below full evaluation decided the same way (−0.21 [−0.47, −0.01]). The evidence’s standard errors are wide relative to candidates’ effects, so whether a rule gains or loses is decided mostly by its within-band acceptances.

Calibration of the bands. Table 24 gives, for each evidence type at 40 episodes, the share of evidence draws whose interval, the estimate $\pm$ twice its standard error, covers the candidate’s full-evaluation ΔS on the evolve set, which the evidence estimates, and its held-out deployment gain. Paired samples and net evidence are close to their nominal 95% on τ^2 and below it on AppWorld (0.81 and 0.70). Replay—null bands cover their target in only 39–62% of draws, because replays of failures cannot see harm to solved tasks: the shortfall is bias, not variance, which is why replay—null@40 accepted the change that blocked legitimate writes. Full evaluation’s own band covers the deployment gain in 90–100% of candidates.

The loop’s whole cost. Table 25 prices the RRSI loop itself, not this study’s extra evidence or deployments: the proposal side’s model calls (analyst, digester, proposer, critic), the candidates’ full evaluations, smoke tests and the base-harness calibration, at list prices. Candidate evaluation is 24–54% of it on τ^2 and 45% on AppWorld; the episodes sequential full evaluation would have saved are 19% of the loop’s cost on r1–r3 and 17% on AppWorld. With a thinking proposer (r4) the model calls are 93% of the cost and sequential evaluation saves 1%.

What each rule buys. Every rule uses what the loop already has: the incumbent’s full evaluation, which it ran when it accepted the incumbent, and its failed trials. Sample and sequential rules buy only candidate episodes, paired with the incumbent’s existing trials. Replay rules buy candidate and null replays from recorded failures; null replays are shared by a round’s candidates. Net evidence buys replays and candidate runs on solved tasks. None of them needs anything the next round would otherwise lack: an accepted candidate’s evaluation becomes the next incumbent’s, so under sequential evaluation accepted candidates complete their evaluation, and only the scores of rejected

Rule	r2+r3: non-inferiority to full evaluation					Agree with full evaluation				p at margin 0.25		
	– full [90%]	0.1	0.25	0.5	0.75	r2+r3	r1	r4	E3	r1	r4	E3
Sequential full	–0.02 [–0.12, +0.07]	0.086	<0.001	<0.001	<0.001	0.93	0.96	0.95	0.98	0.111	0.023	<0.001
Sequential sample@80	+0.03 [–0.26, +0.35]	0.233	0.056	0.002	<0.001	0.78	0.88	0.96	0.89	0.166	0.005	0.136
Sample@80	+0.00 [–0.44, +0.44]	0.330	0.165	0.029	0.004	0.78	0.84	1.00	0.87	0.269	<0.001	0.131
Sample@40	–0.10 [–0.57, +0.35]	0.477	0.280	0.083	0.018	0.67	0.78	0.86	0.78	0.355	0.135	0.170
Net@40	–0.14 [–0.58, +0.27]	0.536	0.309	0.080	0.015	0.63	0.74	0.82	0.68	0.485	0.316	0.264
Replay–null@40	–0.43 [–0.94, +0.05]	0.868	0.728	0.392	0.141	0.68	0.81	0.77	0.69	0.565	0.182	0.113
Keep incumbent	–0.04 [–0.43, +0.35]	0.390	0.170	0.028	0.004	0.74	0.88	0.80	0.85	0.433	0.778	0.202

Table 22: Non-inferiority to full evaluation at smaller margins (one-sided bootstrap p , block bootstrap as in Table 2, not adjusted; 0.75 is the registered margin), and the expected share of rounds in which a rule makes full evaluation’s choice. For keeping the incumbent, Agree is the share of rounds in which full evaluation keeps it. r1 is the exploratory set of the sequential rules; r4 has ten airline rounds and three blocks.

Rule	Registered rule		Full evaluation with own band		One criterion, no within-band branch	
	D [90%]	Accepts	D [90%]	Accepts	D [90%]	Accepts
Full (RRSI) / own band	+0.04 [–0.35, +0.43]	0.26	+0.03 [–0.47, +0.54]	0.36	+0.09 [–0.06, +0.36]	0.03
Sequential full / own band	+0.02 [–0.34, +0.35]	0.20	+0.08 [–0.21, +0.41]	0.20	+0.09 [–0.05, +0.35]	0.02
Sequential sample@80	+0.07 [–0.26, +0.41]	0.23	+0.07 [–0.26, +0.41]	0.23	+0.10 [–0.02, +0.35]	0.02
Sample@80	+0.04 [–0.46, +0.57]	0.38	+0.04 [–0.46, +0.57]	0.38	+0.10 [–0.02, +0.36]	0.02
Sample@40	–0.06 [–0.58, +0.45]	0.40	–0.06 [–0.58, +0.45]	0.40	+0.04 [–0.04, +0.17]	0.02
Net@40	–0.10 [–0.62, +0.39]	0.44	–0.10 [–0.62, +0.39]	0.44	–0.00 [–0.04, +0.04]	0.02
Replay–null@40	–0.39 [–0.79, –0.03]	0.20	–0.39 [–0.79, –0.03]	0.20	–0.12 [–0.30, +0.03]	0.07
LLM judge	–0.40 [–1.41, +0.67]	0.81	–0.40 [–1.41, +0.67]	0.81	–0.40 [–1.41, +0.67]	0.81
None	–0.47 [–1.52, +0.62]	0.91	–0.47 [–1.52, +0.62]	0.91	–0.47 [–1.52, +0.62]	0.91

Table 23: Decision value on r2+r3 (common error check, points per round, block bootstrap) under three decision rules. “Own band”: full and sequential full evaluation decided with twice their own standard error, as the other rules already are, so only their rows change. “One criterion”: every rule admits a candidate only if its estimate exceeds twice its standard error and the cost condition holds.

Evidence (@40)	r1	r2+r3	r4	E3
Sample@40	0.98 / 0.94	0.98 / 0.92	1.00 / 0.96	0.81 / 0.80
Net@40	0.95 / 0.94	0.92 / 0.94	0.99 / 0.95	0.70 / 0.83
Replay–null@40	0.54 / 0.76	0.51 / 0.66	0.62 / 0.88	0.39 / 0.67
Replay@40	0.12 / 0.15	0.16 / 0.15	0.32 / 0.00	0.24 / 0.45
Full evaluation	1.00 / 0.94	1.00 / 0.90	1.00 / 1.00	1.00 / 0.97

Table 24: Coverage of ± 2 standard-error intervals: evolve ΔS / deployment gain, averaged over candidates (200 evidence draws each). Full evaluation covers its own ΔS by construction.

Trajectory	Loop	Model	Cand.	Smoke	Sequential full saves	
	\$	calls %	eval. %	calib. %	episodes	% of loop
r1 airline	6.52	69	27	4	793	15
r1 retail	10.73	47	46	6	1,536	20
r2 airline	6.92	64	32	4	437	8
r2 retail	11.92	41	53	6	2,361	32
r3 airline	7.10	72	24	3	392	7
r3 retail	11.83	40	54	6	1,604	21
r4 airline	9.39	93	5	2	93	1
E3 AppWorld	18.58	52	45	3	1,318	17

Table 25: Dollar cost of the RRSI loop per trajectory and domain (r4 retail, stopped after two rounds, omitted), its shares, and the evolve episodes sequential full evaluation would have saved, priced at the mean price of a candidate-evaluation episode.

candidates, which RRSI shows its proposer, become early-stopped estimates. The fixed-budget rules would need an extra full evaluation of each accepted candidate to give the next round the same traces; our counts omit it, which favours them.

Change cost. Net(N) sets $\Delta C_{\text{change}} = 0$ because every rule sees the same candidates, a simplification: reviewing and maintaining an accepted change costs something, and rules accept different numbers of changes. Charging κ episodes per accepted change lowers a rule’s Net(10^4) by κ times its acceptance rate. On r2+r3 at $v = 10$, sequential sampling, sequential full evaluation and full evalua-

tion have +10 [–315, +350], –71 [–438, +254] and –108 [–500, +274] episodes at $\kappa = 0$ and –215, –267 and –370 at $\kappa = 1,000$; the fixed-budget rules, which accept 38–44% of rounds, fall further.

Registration timeline. All times UTC, 2026, from the repository history. 7 Oct 11:26: r1’s registration, analysis code and launch script committed; the loop and its held-out deployments started after

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it. 15:34–15:54: the registered power rule raised held-out trials (Appendix N). 20:22: r1 finished; 20:23 its final disagreement rate logged. 8 Oct 00:24: r2 and r3 registered, before any of their episodes; 00:45: addendum (thinking roles for r4, verify-one-candidate rules). 01:00: E3 registered, before any episode of its loop. 02:06: the sequential rules and their tests registered for r2, r3 and E3 (addendum 2); no decision value of r2, r3 or E3 had been computed. 08:01: r2 and r3’s analysis code committed, tested on r1 only. 09:55: every round of r2 and r3 deployed; 10:00–10:15 their registered analyses run as committed. 16:32 and 18:02: E3’s loop and deployments finished; 18:14 its registered analysis run. The operations logs in the registration files record the restarts, the r4 truncation and the held-out rates seen in progress logs.

R Reproducibility

The code, environments, banks and Study 2’s registered pipeline are submitted as anonymised supplementary material. *Study 1*: the environments and fault suites (agent_exp/env.py, agent_exp/tau2_env.py), every prompt used for attribution, taxonomy induction, labelling, patch generation and judging (agent_exp/analyst.py), the agent and analyst settings (DeepSeek-V4-Flash and DeepSeek-V4-Pro through the provider’s Anthropic-compatible endpoint, at most 4,000 output tokens per agent call and 8,000 per structured analyst call, provider-default sampling), and the complete intervention banks (runs/tau2_*/bank.json: every replay, judge check, null replay and regression run with its token count and source trace). Bank construction is deterministic given the seeds in agent_exp/bank.py. Every table and figure of Study 1 is produced offline from the banks and held-out runs by agent_exp.report_tau2, from the outputs of agent_exp.reanalysis, reanalysis3, reanalysis5 (binomial harm model), deployment (deployment gains, $\text{Net}(N)$, transfer) and judge_configs. *Study 2*: the vendored RRSI loop with a list of our changes (rrsi/, rrsi/NOTICE.md), the τ^2 domains (domains/tau2*), the harness check against τ^2 ’s orchestrator (agent_exp/harness_check.py), and the evidence pipeline, pre-registration and registered analysis (verify/); Table 20 is produced by verify.report_e1 from the output of verify.analyze, the analyses with the common

error check by verify.posthoc_e1, and r2 and r3’s registered analysis and Table 2 by verify.e1r and verify.report_e1r. Re-collecting any run needs API access and is not bit-reproducible, because the provider does not offer seeded sampling.

S The CARVE Design Study

Why the simulator was optimistic. Our first design study, CARVE, used a simulator in which it needed half the budget of the strongest baseline (Appendix S). Calibrated to the measured judge accuracy, spurious recovery, analyst accuracy and cost ratio, the simulator still predicts that CARVE leads uncertainty sampling by 14 points on retail at budget 50; on the bank it trails by 15 points at budget 40 (significant harm, $c_w = 0.02$). The simulator shared the method’s structure: one b for all categories, value spread over many edges, and a fixed cost c that made every wrong change harmful, an assumption that was never measured and on which the value of verification largely depends.

Our first approach treated judge checks as cheap and replays as expensive, and allocated a budget over both by Bayesian experimental design. We keep it here as the multi-fidelity method compared in §4, together with the simulator in which it was developed. The simulator shares the method’s structure. Its results show the method working under its own assumptions, and they are not evidence about real agents.

Calibrated simulation. With the measured judge accuracy, spurious recovery, analyst accuracy and cost ratio, the simulator predicts that CARVE runs no judge checks on ShopDesk and retail and only a few on airline, which matches the break-even analysis in Table 4. For τ^2 retail it predicts net gains at budget 50 of 0.51 for CARVE and 0.38 for uncertainty sampling; for airline, 0.33 and 0.31; for ShopDesk, 0.81 and 0.79.

S.1 Problem Formulation

An agent is built from *components* $j \in \{1, \dots, J\}$: sections of its system prompt, tool descriptions, tool implementations, orchestration settings. Its failures have been grouped into *categories* $i \in \{1, \dots, I\}$ by an LLM, with f_i the share of failures in category i . For every cell (i, j) an analyst LLM can propose K candidate patches a_{ijk} that rewrite component j to address category i .

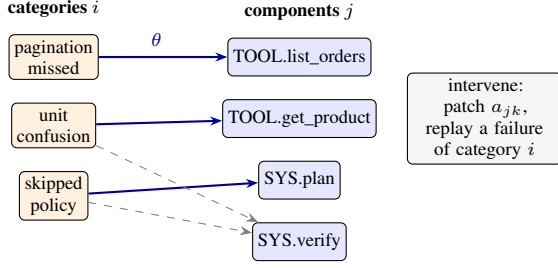


Figure 2: The category–component graph. Solid edges are true causal edges; dashed ones are plausible but wrong attributions (e.g. blaming verification for a unit error). Each intervention patches one component and replays one failure.

Interventions. An intervention (i, j, k, ϕ) samples a failed trajectory labelled i , applies patch a_{ijk} , and observes a binary outcome at fidelity ϕ . A *full replay* ($\phi = F$) resumes the episode from the attributed failing step and records whether the task now succeeds. A *single-step check* ($\phi = S$) regenerates only the failing step and asks a judge whether the diagnosed error is gone. They cost c_F and $c_S \ll c_F$.

Edges and patch quality. We write $e_{ij} \in \{0, 1\}$ for whether component j is a cause of category i , meaning some change to j repairs a non-trivial fraction of its failures, and $q_{ijk} \in [0, 1]$ for the repair probability of patch k when $e_{ij} = 1$. This separation matters because a negative replay outcome is ambiguous: either j is the wrong component or a_{ijk} is a bad patch. Testing several patches per cell resolves the ambiguity.

Objective. After spending at most budget B , the method accepts a set of cells $\mathcal{A}$ and, for each, a patch k_{ij}^* . Accepting a true edge repairs a fraction q_{ijk^*} of category i ’s failures; accepting a false edge costs a regression c_{fp} , reflecting that unnecessary changes perturb behaviour on tasks that used to succeed (Vasudev et al., 2026). We measure

$$G(\mathcal{A}) = \sum_i f_i \left(1 - \prod_{j \in \mathcal{A}_i^+} (1 - q_{ijk_{ij}^*}) \right) - c_{fp} |\{(i, j) \in \mathcal{A} : e_{ij} = 0\}|, \quad (5)$$

where $\mathcal{A}_i^+ = \{j : (i, j) \in \mathcal{A}, e_{ij} = 1\}$. This is the net failure mass repaired; we report it as a fraction of the oracle value obtained by accepting exactly the true edges with their best patches.

S.2 CARVE

Observation model For a trajectory labelled i and patch a_{ijk} , the probability that a full replay succeeds is

$$p_{ijk} = \begin{cases} \lambda q_{ijk} + (1 - \lambda)b & e_{ij} = 1, \\ b & e_{ij} = 0, \end{cases} \quad (6)$$

where b is the spurious recovery rate (stochastic re-generation sometimes succeeds regardless of the patch) and λ is the accuracy of the category labels used to sample trajectories; a mislabelled trajectory is treated as unaffected by the patch. A single-step check returns $z = 1$ with probability $sp + \epsilon(1 - p)$, where s and ϵ are the sensitivity and false-positive rate of the check relative to the full outcome. All four nuisance parameters can be estimated from a small calibration set on which both fidelities and a human audit are run.

Attribution-informed prior Let n_{ij} be the number of failures in category i that the analyst attributed to component j , and $\hat{r}$ the analyst’s attribution accuracy estimated on a small audited sample. With d the expected number of causal components per category and $\rho = d/J$ the base edge density, we set

$$\pi_{ij} = (1 - \hat{r})\rho + \hat{r} \min(1, dn_{ij}/n_i), \quad (7)$$

so that an unreliable analyst ($\hat{r} \rightarrow 0$) yields the flat prior ρ and a reliable one concentrates mass on the components it names. Eq. 7 is a heuristic interpolation, not a calibrated model; with a larger audit, π could instead be fitted by logistic regression on audited attributions. Patch quality has prior $q_{ijk} \sim \text{Beta}(\alpha_q, \beta_q)$.

Posterior Cells are independent given the data. For each cell we keep the posterior log-odds of e_{ij} and, for each patch, a posterior over q_{ijk} on a fixed grid of $G = 101$ points. With $\mathcal{D}_{ijk}$ the outcomes observed for patch k ,

$$\text{logit } P(e_{ij}=1 \mid \mathcal{D}) = \text{logit } \pi_{ij} + \sum_k \log(\bar{L}_{ijk}^1 / L_{ijk}^0), \quad (8)$$

where $\bar{L}_{ijk}^1 = \int P(\mathcal{D}_{ijk} \mid q, e_{ij}=1) dP(q)$ is the marginal likelihood of patch k ’s outcomes under a true edge and $L_{ijk}^0 = P(\mathcal{D}_{ijk} \mid e_{ij}=0)$. Every update multiplies one grid vector by a likelihood vector, so updates and look-aheads are vectorised dot products.

Selecting interventions The value of the current decision for cell (i, j) , with P_{ij} its edge posterior and $\hat{Q}_{ij} = \max_k \mathbb{E}[q_{ijk} \mid \mathcal{D}]$, is

$$V_{ij} = \max(P_{ij} f_i \hat{Q}_{ij} - (1 - P_{ij}) c_{fp}, 0), \quad (9)$$

the expected contribution of the cell to Eq. 5 under the Bayes decision (accept iff the first term is positive), treating cells as additive. Eq. 5 combines components of one category multiplicatively, so V_{ij} slightly overstates the value of a second cause for an already repaired category; we accept this approximation because it keeps cells independent. For an action $a = (i, j, k, \phi)$ repeated n times, let $V'_{ij}(s)$ be the value after observing s successes. The n -step knowledge gradient is

$$\text{KG}_n(a) = \sum_{s=0}^n P(s \mid \mathcal{D}, a) V'_{ij}(s) - V_{ij}, \quad (10)$$

and CARVE runs the action maximising

$$\text{score}(a) = \max_{n \in \{1, 2, 4, 8\}} \frac{\text{KG}_n(a)}{n c_\phi}. \quad (11)$$

Looking ahead several observations matters because a single binary outcome often cannot flip a decision, in which case $\text{KG}_1 = 0$ even for a valuable cell; maximising over n is the standard remedy (Frazier et al., 2008). Because cells are independent, only the observed cell’s scores change after each outcome, so each step rescores one cell in $O(KG)$ time after an initial pass over all cells.

Decisions and stopping. At any point the method accepts the cells whose current value is positive and applies, for each, the patch with the highest posterior mean quality. It stops early when no action has a score above a small threshold, i.e. when no remaining experiment is expected to be worth its cost.

S.3 When Do Priors and Cheap Checks Help?

These results analyse an idealised per-cell sequential test, not CARVE’s full policy; they explain which quantities govern the method’s behaviour. We consider a single cell and a single patch with known success probabilities p_1 (true edge) and $p_0 = b$, under a rule that stops when the posterior log-odds leaves $[-A, A]$, $A = \log \frac{1-\delta}{\delta}$. Let $D_1 = \text{KL}(p_1 \parallel p_0)$ and $D_0 = \text{KL}(p_0 \parallel p_1)$ be the expected log-likelihood-ratio drift per observation under each hypothesis. Proofs, which follow Wald’s approximations (Wald, 1945), are in Appendix S.5.

Proposition 3 (Value of the LLM prior). *Let $|\ell| < A$ and neglect overshoot and stopping at the wrong boundary (whose probability is approximately $\delta e^{-\ell}$ for a true edge and δe^ℓ for a false one). A cell with initial log-odds ℓ then needs $N_1(\ell) = (A - \ell)/D_1$ observations in expectation if it is a true edge and $N_0(\ell) = (A + \ell)/D_0$ if not. Replacing a flat prior ℓ_ρ by LLM-derived log-odds ℓ_{ij} saves, in expectation over cells,*

$$\Delta = \frac{\rho \mathbb{E}[\delta \ell \mid e=1]}{D_1} - \frac{(1 - \rho) \mathbb{E}[\delta \ell \mid e=0]}{D_0}, \quad (12)$$

with $\delta \ell = \ell_{ij} - \ell_\rho$, observations per cell, which is positive exactly when the prior raises true edges more, relative to their base rate and drift, than it raises false ones.

Under Eq. 7, $\ell_{ij} \rightarrow \ell_\rho$ as $\hat{r} \rightarrow 0$, so $\Delta \rightarrow 0$: a useless analyst costs nothing beyond the flat prior, whereas trusting its labels outright (LLM-only) makes permanent errors. A miscalibrated analyst can make Δ negative, but only by an amount linear in the log-odds shift, after which the evidence takes over. Appendix S.7 compares these predictions with simulated sequential tests.

Proposition 4 (Cheap checks). *Let D_e^S denote the drifts of single-step outcomes, $z \sim \text{Bern}(s p_e + \epsilon(1 - p_e))$. Under the same approximations, a test using only single-step checks reaches the stopping boundary at lower expected cost than one using only full replays if and only if*

$$\frac{c_S}{c_F} < \frac{D_e^S}{D_e}, \quad (13)$$

where e is the true state of the cell.

Since e is unknown, the proposition gives a clear rule only when the cost ratio lies outside the interval spanned by the two critical ratios; inside it, which fidelity is better depends on the posterior, which is what CARVE’s look-ahead weighs.

With our default parameters ($p_1 = 0.54$, $b = 0.05$, $s = 0.85$, $\epsilon = 0.15$) the critical ratio is 0.31 for true edges and 0.44 for false ones; with a weak check ($s = 0.65$, $\epsilon = 0.35$) it falls to 0.05–0.08. Section S.4 tests both predictions.

S.4 Controlled Experiments

Real agents do not reveal which component truly caused a failure, so we first evaluate in a testbed where the causal graph is known and the method’s model is deliberately misspecified.

Setup

Testbed. Each world has $I = 12$ categories with Zipfian frequencies, $J = 20$ components and $K = 3$ patches per cell. Every category has one or two true components and one *decoy* component that wrong attributions prefer, mimicking symptom-level mislabelling. Patch quality is bimodal: with probability 0.3 a patch is bad ($q \sim \text{Beta}(1, 9)$), otherwise $q \sim \text{Beta}(6, 3)$. A trajectory sampled for category i is mislabelled with probability $1 - \lambda = 0.2$, in which case it comes from another category (weighted by frequency) and is repaired only if *its own* edge to j exists, unlike the model’s flat treatment of mislabelled traces. The analyst attributes 20 failures per category with accuracy $r = 0.5$; wrong attributions go to the decoy with probability 0.6. The analyst’s accuracy is estimated from an audit of 20 attributions. Costs are $c_F = 1$, $c_S = 0.1$, $c_{fp} = 0.02$, and $s = 0.85$, $\epsilon = 0.15$, $b = 0.05$. Section S.4 relaxes the assumptions the generator shares with the model. Full details are in Appendix S.6.

Baselines. *LLM-only* accepts, for every category, the most-attributed component and any component holding at least 40% of the attributions, with the first patch, i.e. it acts on an induced taxonomy without verification. *Replay-each* tests attributed hypotheses in order of attribution count with three full replays of the first patch and accepts a hypothesis if at least two succeed, mirroring do-then-verify methods such as DoVer; *Replay-each, 3 patches* gives each hypothesis two replays of each of three patches and accepts if any patch succeeds twice; *Sequential-each* replays each hypothesis, cycling through patches, until CARVE’s own posterior leaves $[0.05, 0.95]$. Untested hypotheses fall back to LLM-only, except for *Sequential-each*, which uses the Bayes decision. *Thompson sampling* samples (e, q) from the posterior and replays the cell and patch with the largest sampled repair. *Uncertainty sampling* replays the cell with the largest frequency-weighted uncertainty $f_i \min(P_{ij}, 1 - P_{ij})$, cycling through patches, at full fidelity; *Uncertainty + cheap checks* first runs three single-step checks per (cell, patch) and then confirms with full replays. All baselines except LLM-only and the *Replay-each* variants share CARVE’s posterior and decision rule, so comparisons among them isolate the selection rule.

Method	Budget (full-replay units)			
	20	50	100	200
LLM attribution only	0.297 $\pm$ 0.021	0.297 $\pm$ 0.021	0.297 $\pm$ 0.021	0.297 $\pm$ 0.021
Replay-each (do-then-verify)	0.255 $\pm$ 0.021	0.270 $\pm$ 0.021	0.466 $\pm$ 0.027	0.468 $\pm$ 0.026
Replay-each, 3 patches	0.301 $\pm$ 0.025	0.317 $\pm$ 0.026	0.401 $\pm$ 0.026	0.612 $\pm$ 0.026
Sequential-each (adaptive stopping)	0.426 $\pm$ 0.016	0.522 $\pm$ 0.018	0.643 $\pm$ 0.016	0.813 $\pm$ 0.014
Thompson sampling	0.407 $\pm$ 0.020	0.419 $\pm$ 0.020	0.500 $\pm$ 0.017	0.625 $\pm$ 0.014
Uncertainty sampling	0.455 $\pm$ 0.016	0.581 $\pm$ 0.019	0.690 $\pm$ 0.017	0.804 $\pm$ 0.016
Uncertainty + cheap checks	0.549 $\pm$ 0.018	0.662 $\pm$ 0.016	0.784 $\pm$ 0.015	0.876 $\pm$ 0.009
CARVE (all components)	0.666 $\pm$ 0.019	0.787 $\pm$ 0.015	0.853 $\pm$ 0.011	0.902 $\pm$ 0.009
full replay only	0.514 $\pm$ 0.016	0.609 $\pm$ 0.020	0.697 $\pm$ 0.017	0.825 $\pm$ 0.012
myopic ($n=1$ look-ahead)	0.591 $\pm$ 0.016	0.652 $\pm$ 0.014	0.706 $\pm$ 0.016	0.729 $\pm$ 0.014
uninformed prior	0.347 $\pm$ 0.019	0.474 $\pm$ 0.017	0.631 $\pm$ 0.015	0.800 $\pm$ 0.012
one patch per cell	0.571 $\pm$ 0.019	0.636 $\pm$ 0.020	0.673 $\pm$ 0.019	0.688 $\pm$ 0.020
no label-noise term	0.663 $\pm$ 0.020	0.772 $\pm$ 0.018	0.864 $\pm$ 0.012	0.922 $\pm$ 0.006

Table 26: Net gain as a fraction of the oracle (mean $\pm$ s.e., 50 worlds). The lower block ablates CARVE.

Metrics. Net gain (Eq. 5) as a fraction of the oracle, and edge F1. We report means and standard errors over 50 worlds for the main experiment and 30 for the others; all methods see the same worlds, so differences are paired.

Main results Figure 3 and Table 26 show that CARVE is best at every budget. With 50 units it recovers 79% of the oracle gain; the strongest baseline, uncertainty sampling with the same cheap checks, recovers 66% and needs about twice the budget (100 units, 78%) to match it. The gap narrows at large budgets (90% vs. 88% at 200), as every reasonable method eventually resolves the frequent categories.

Where the gain comes from. Two comparisons separate the selection rule from the cheap checks. With full replays only, CARVE beats uncertainty sampling by 1–6 points (e.g. 0.609 vs. 0.581 at budget 50): the knowledge-gradient rule alone helps only modestly. Given cheap checks, its advantage grows to 7–13 points (0.787 vs. 0.662 at 50), because it decides per action whether a check is informative enough for its price and stops checking a cell once checks can no longer change the decision, whereas a fixed check-then-confirm schedule wastes checks on resolved cells. Most of CARVE’s advantage is therefore in *how it uses* cheap checks, which also makes their calibration critical (§S.4).

Verification baselines. LLM-only is stuck at 30% because it accepts decoys and bad first patches. *Replay-each* is initially *worse* than not verifying: with $p_1 \approx 0.54$, requiring two successes in three replays rejects about 44% of true edges even when the patch is good, and its fixed per-hypothesis cost exhausts small budgets on frequent but already-certain hypotheses. Testing three patches (61% at 200) or stopping adaptively (81% at 200) helps, but both remain well below CARVE at small bud-

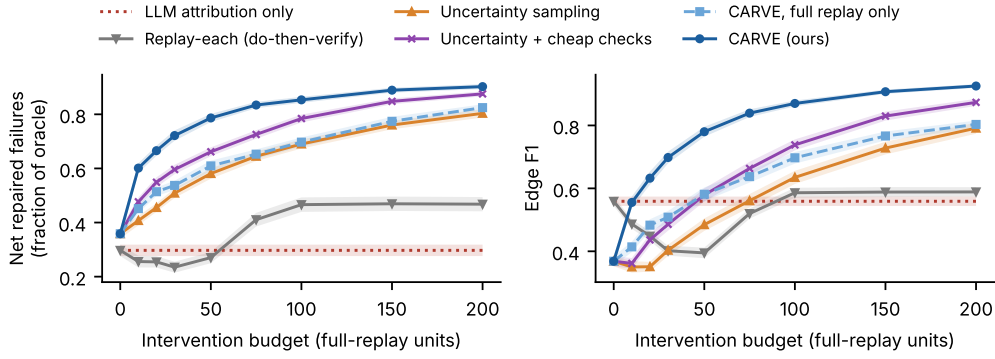


Figure 3: Net repaired failure mass (left) and edge F1 (right) against intervention budget, 50 worlds, shaded ± 1 s.e.

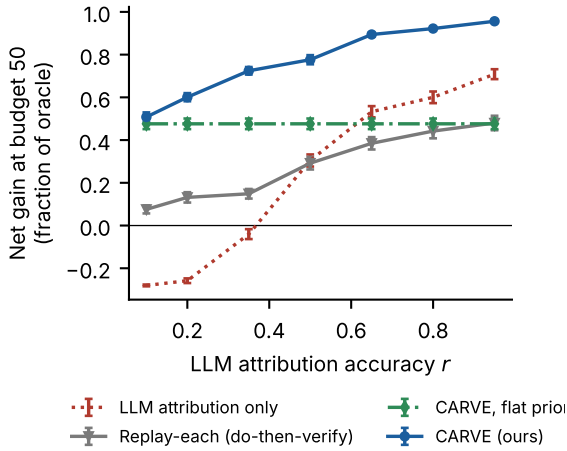


Figure 4: Net gain at budget 50 as the analyst’s attribution accuracy r varies (30 worlds, ± 1 s.e.).

gets (32% and 52% at 50). Thompson sampling spends most replays on cells it already believes in and reaches only 63% at 200.

Ablations. Removing single-step checks costs 15–18 points at small budgets. Restricting the look-ahead to one observation is below the full method at every budget and plateaus at 73%, because many cells cannot be flipped by a single outcome and are never revisited. A flat prior costs 32 points at budget 20 and still 10 points at 200. With one patch per cell the method plateaus at 69%: it cannot tell a wrong component from a bad patch and rejects true edges for good. Dropping the label-noise term makes no difference at the default label accuracy $\lambda = 0.8$ (it is even slightly better at large budgets, 0.922 vs. 0.902 at 200), but it matters when labels are worse (§S.4).

When does the LLM prior help? Figure 4 varies the analyst’s accuracy r . As Proposition 3

suggests, CARVE improves monotonically with r and is never below the flat-prior variant: at $r = 0.1$ it scores 0.51 ± 0.02 against 0.48 ± 0.03 . The prior is not exactly flat there (the decoy still receives about twice the base rate), but its shifts are small, so evidence quickly dominates. Acting on attributions directly is harmful below $r \approx 0.35$, where LLM-only has a *negative* net gain (regressions outweigh repairs), and even at $r = 0.95$ it reaches only 71% of the oracle against 96% for CARVE, since it cannot detect bad patches. Replay-each improves with r and only matches the flat-prior variant of CARVE at $r = 0.95$ (0.48 vs. 0.48).

When do cheap checks help? Figure 6 (Appendix S.8) is qualitatively consistent with Proposition 4. With the default check, the benefit of cheap checks shrinks as their cost grows and vanishes between 0.35 and 0.5 (0.64 ± 0.02 vs. 0.61 ± 0.03 at 0.35, overlapping; 0.59 vs. 0.61 at 0.5), in line with the predicted critical ratios of 0.31 and 0.44. At cost 0.1, a check of quality $s - \epsilon = 0.5$ (critical ratio 0.14–0.22) still helps, while one of quality 0.3 (critical ratio 0.05–0.08) brings no gain. At cost 0.5 CARVE is slightly below its full-replay-only variant although its action set contains every full-replay action, which shows that the greedy rule is not exactly cost-optimal when the two fidelities are nearly equally efficient.

Robustness The generator shares several assumptions with the model: check errors depend only on the full outcome, cells are independent, the analyst’s accuracy is estimated without bias, and the regression cost is small. Table 27 and Appendix S.8 relax each in turn.

Worse labels. With label accuracy $\lambda = 0.5$ both methods lose ground, and the label-noise term now

Condition (budget 50)	Unc. + checks	CARVE
Default testbed	0.656 $\pm$ 0.022	0.776 $\pm$ 0.021
Label accuracy $\lambda = 0.5$	0.513 $\pm$ 0.023	0.688 $\pm$ 0.023
Analyst overconfident ($\hat{r}+0.3$)	0.752 $\pm$ 0.018	0.857 $\pm$ 0.014
Analyst underconfident ($\hat{r}-0.3$)	0.495 $\pm$ 0.023	0.653 $\pm$ 0.032
Judge fooled by bad patches	0.579 $\pm$ 0.022	0.377 $\pm$ 0.027
Regression cost $c_{fp} = 0.005$	0.711 $\pm$ 0.022	0.798 $\pm$ 0.016
Regression cost $c_{fp} = 0.05$	0.643 $\pm$ 0.024	0.793 $\pm$ 0.019
Regression cost $c_{fp} = 0.1$	0.624 $\pm$ 0.023	0.823 $\pm$ 0.018

Table 27: Violations of the model’s assumptions (30 worlds). The judge-fooled row is the one setting where CARVE loses.

matters: CARVE scores 0.688, against 0.588 without the term and 0.513 for the baseline.

Biased analyst audit. If the audit overstates the analyst’s accuracy by 0.3, CARVE actually improves (0.857), because a stronger prior concentrates the budget and the evidence corrects the decoys. Understating it by 0.3 costs 12 points (0.653), still above the baseline. Under-trusting a decent analyst is thus the more costly error in this testbed.

Regression cost. Raising c_{fp} from 0.005 to 0.1 leaves CARVE between 0.78 and 0.82, while LLM-only falls from 0.39 to -0.13 : the more costly a wrong change, the more verification is worth.

Joint causes. When 30% of categories need two components changed together (a single patch then repairs only a fifth as often), violating cell independence, CARVE drops from 0.787 to 0.661 at budget 50 but stays ahead of uncertainty sampling with checks (0.558; Table 30).

Correlated check errors. The one setting where CARVE loses is when the single-step judge is fooled by plausible-looking bad patches (false-positive rate 0.6 for patches with $q < 0.3$). CARVE then trusts checks that systematically pass bad patches and falls to 0.377, below every method that relies on full replays (0.58–0.61). Modelling the dependence (a quality-dependent false-positive rate in the likelihood) restores it to 0.584, on par with full replays, at a cost of 9 points when the judge is not fooled (0.684 vs. 0.776). Whether real judges behave this way is therefore the most important quantity to measure before relying on cheap checks.

Nuisance parameters and scale. Misspecifying b , λ or the patch-quality prior changes the result by at most 5 points (Table 31, Appendix S.8); misstating the check’s sensitivity and false-positive rate costs 6–14 points. On a larger graph ($I = 20$,

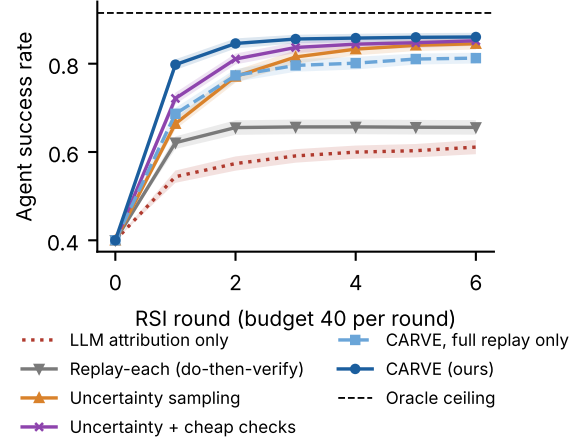


Figure 5: Agent success over six rounds of attribute-verify-patch, with a budget of 40 units per round (30 worlds, ± 1 s.e.).

$J = 50$; Table 29) the gap widens: at budget 50 CARVE reaches 78% of the oracle against 60% for uncertainty sampling with checks.

Inside a self-improvement loop Finally we embed each method in a loop. The agent starts at 40% success. In each round the analyst attributes 240 new failures, sampled in proportion to the current failure distribution; the method spends 40 units; accepted patches are applied, which repairs failures (true edges) or adds a regression of c_{fp} times the current failure mass (false edges); and the failure distribution is recomputed, so priorities shift towards categories that remain. Posteriors carry over between rounds.

Figure 5 shows that CARVE reaches 80% success after one round and 85% after two, close to its plateau of 86% (the oracle ceiling is 91.5%). Uncertainty sampling with checks needs four rounds to reach 84% and ends at 85%; without checks it needs six. LLM-only and Replay-each plateau at 61% and 66%: once their accepted false edges and rejected true edges are fixed in place, new attributions do not correct them.

S.5 Proofs for CARVE’s propositions

Proposition 3. Let $L_t = \ell + \sum_{u \leq t} \log \frac{P(o_u | e=1)}{P(o_u | e=0)}$ be the posterior log-odds after t observations. Under $e = 1$, the increments are i.i.d. with mean $D_1 = \text{KL}(p_1 \| p_0) > 0$. The rule stops at $N = \inf\{t : L_t \notin (-A, A)\}$. By Wald’s identity, $\mathbb{E}_1[L_N - \ell] = D_1 \mathbb{E}_1[N]$. Wald’s approximations give $P_1(L_N \leq -A) \approx e^{-(A+\ell)} \approx \delta e^{-\ell}$

and $P_0(L_N \geq A) \approx \delta e^\ell$. Neglecting overshoot and this probability (which requires $\delta e^{|\ell|} \ll 1$), $L_N = A$, hence $\mathbb{E}_1[N] = (A - \ell)/D_1$; if $\ell \geq A$ no observation is taken, which is why we assume $|\ell| < A$. Symmetrically, under $e = 0$ the drift is $-D_0$ and $\mathbb{E}_0[N] = (A + \ell)/D_0$. Taking the difference between ℓ_ρ and ℓ_{ij} and averaging over cells with $P(e=1) = \rho$ gives Δ . When the patch quality q is unknown, outcomes are exchangeable rather than i.i.d. under the marginal likelihood, and Wald’s identity does not apply directly; conditioning on q recovers the result with $D_1 = D_1(q)$, which approaches zero for a bad patch. This is the formal reason a single bad patch can stall verification and why CARVE keeps several patches per cell.

Proposition 4. By the same argument a single-step test needs $(A \mp \ell)/D_e^S$ observations at cost c_S each, and a full-replay test needs $(A \mp \ell)/D_e$ at cost c_F . The single-step test is cheaper iff $c_S/D_e^S < c_F/D_e$. Since the single-step outcome is a garbling of the full outcome, the data-processing inequality gives $D_e^S \leq D_e$, so the critical ratio is at most one.

S.6 Simulation Details

Category frequencies are $f_i \propto 1/i$. For each category the number of true components is uniform on $\{1, 2\}$ and the decoy is a further distinct component. The audit estimate $\hat{r}$ is a binomial proportion from 20 audited attributions. The prior uses $d = 1.5$. The model’s grid has 101 points; look-ahead horizons are $\{1, 2, 4, 8\}$; the stopping threshold is 10^{-6} . Replay-each uses three replays and accepts with at least two successes. Uncertainty sampling switches to an untried patch once the current patch has three full observations. In the loop experiment, 240 attributions per round are allocated to categories by a multinomial draw with the current failure distribution. All randomness is seeded; the code reproduces every number in this paper with `python -m sim.experiments all`.

S.7 Checking Proposition 1

Table 28 compares the predicted expected number of observations with simulated sequential tests ($p_1 = 0.543$, $p_0 = 0.05$, $\delta = 0.02$, 4000 runs per row). The predictions capture the direction and approximate size of the effect of the initial log-odds; they are least accurate when $|\ell|$ is large, where overshoot and wrong-boundary exits (last column) are no longer negligible.

ℓ	e	predicted N	simulated N	wrong exit
-3	1	7.17	5.94	0.242
-3	0	1.55	2.54	0.000
-2	1	6.13	6.28	0.117
-2	0	3.28	3.73	0.001
-1	1	5.09	5.79	0.047
-1	0	5.02	5.21	0.003
+0	1	4.05	4.76	0.012
+0	0	6.75	7.28	0.009
+1	1	3.01	4.27	0.006
+1	0	8.48	8.58	0.015
+2	1	1.97	2.97	0.001
+2	0	10.22	10.23	0.068

Table 28: Proposition 3: predicted vs. simulated observations to a decision.

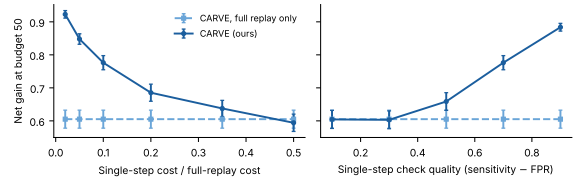


Figure 6: Effect of the single-step check’s cost (left, quality fixed at $s=0.85$, $\epsilon=0.15$) and quality (right, cost fixed at 0.1) on net gain at budget 50.

S.8 Additional Results

Method	50	100	300
LLM attribution only	0.231 $\pm$ 0.033	0.231 $\pm$ 0.033	0.231 $\pm$ 0.033
Replay-each (do-then-verify)	0.224 $\pm$ 0.034	0.400 $\pm$ 0.037	0.531 $\pm$ 0.032
Uncertainty sampling	0.502 $\pm$ 0.022	0.637 $\pm$ 0.020	0.822 $\pm$ 0.015
Uncertainty + cheap checks	0.603 $\pm$ 0.019	0.746 $\pm$ 0.018	0.873 $\pm$ 0.013
CARVE, full replay only	0.613 $\pm$ 0.013	0.710 $\pm$ 0.012	0.834 $\pm$ 0.010
CARVE (ours)	0.780 $\pm$ 0.016	0.860 $\pm$ 0.011	0.933 $\pm$ 0.008

Table 29: Net gain on a larger graph ($I=20$, $J=50$), 20 worlds.

Method	20	50	100	200
LLM attribution only	0.228 $\pm$ 0.028	0.228 $\pm$ 0.028	0.228 $\pm$ 0.028	0.228 $\pm$ 0.028
Replay-each	0.202 $\pm$ 0.027	0.207 $\pm$ 0.023	0.348 $\pm$ 0.033	0.362 $\pm$ 0.036
Thompson sampling	0.325 $\pm$ 0.031	0.343 $\pm$ 0.032	0.398 $\pm$ 0.032	0.484 $\pm$ 0.028
Uncertainty sampling	0.345 $\pm$ 0.036	0.465 $\pm$ 0.035	0.544 $\pm$ 0.035	0.662 $\pm$ 0.029
Uncertainty + cheap checks	0.423 $\pm$ 0.037	0.558 $\pm$ 0.035	0.615 $\pm$ 0.034	0.685 $\pm$ 0.030
CARVE (ours)	0.573 $\pm$ 0.029	0.661 $\pm$ 0.029	0.701 $\pm$ 0.027	0.745 $\pm$ 0.026

Table 30: Net gain when 30% of categories need two components changed together (30 worlds).

Model assumption (true value)	Net gain
All nuisance parameters as in the generator	0.776 $\pm$ 0.021
Spurious rate b : 0.01 (0.05)	0.811 $\pm$ 0.012
Spurious rate b : 0.15 (0.05)	0.826 $\pm$ 0.016
Label accuracy λ : 1.0 (0.8)	0.768 $\pm$ 0.022
Label accuracy λ : 0.6 (0.8)	0.794 $\pm$ 0.017
Check s/ϵ : .95/.05 (.85/.15)	0.717 $\pm$ 0.016
Check s/ϵ : .70/.30 (.85/.15)	0.636 $\pm$ 0.021
Patch prior Beta(1, 1) (Beta(2, 2))	0.811 $\pm$ 0.015
Patch prior Beta(5, 2) (Beta(2, 2))	0.777 $\pm$ 0.020

Table 31: CARVE with misspecified nuisance parameters, budget 50, 30 worlds. Several “wrong” values do slightly better than the generator’s, a reminder that the model is misspecified in other ways too.